A Privacy-Preserving Data Generation Framework in Federated Learning for Enhanced IoT System Attack Detection

Nguyen Nam Khanh*, Nguyen Son Truong Giang*, Nguyen Tri Dung*,
Vu Duc Lam*, and Nguyen Trong Hoang Dat*
*Department of Information Technology, FPT University, Hanoi, Vietnam
Email: {khanhnn, giangnst, dungnt, lamvd, datnth}@fpt.edu.vn
†Supervisor: Dr. Nguyen Anh Nhat, Co-supervisor: MSc. Luu Minh Tri
Department of Information Technology, FPT University, Hanoi, Vietnam
Email: {nhatna, triml}@fe.edu.vn

Abstract*—Federated learning (FL) enables collaborative training of intrusion detection systems (IDS) across distributed Internet of Things (IoT) edge gateways without centralizing raw telemetry traces. However, extreme minority attack class imbalance and Non-IID traffic distributions severely degrade downstream classification sensitivity, while gradient exchanges remain vulnerable to gradient inversion and membership inference attacks. Although federated generative data augmentation can alleviate severe class skew, deep generative models (e.g., GANs and VAEs) risk memorizing private telemetry, incur steep communication overhead during parameter aggregation, and exhibit numerical instability under stringent differential privacy (DP) noise. In this paper, we propose PP-FedData, a privacy-preserving federated generative framework tailored for IoT-MQTT protocol telemetry. PP-FedData establishes a rigorous empirical benchmark comparing federated marginal-based synthesis (FedDP-Marginal) against a deep federated Conditional Variational Autoencoder (Fed-CVAE). We formulate client-side privacy via local and distributed differential privacy coupled with cryptographic Secure Aggregation (SecAgg$^+$). Under identical non-IID partitions on the DoS/DDoS-MQTT-IoT dataset, we evaluate the fundamental trade-offs between formal privacy bounds (ε, δ), communication complexity, synthetic sample fidelity, and downstream minority-class intrusion detection performance. Our findings demonstrate that FedDP-Marginal achieves superior minority-class detection on discrete header anomalies under strict privacy ($\varepsilon \leq 1.0$) while reducing communication bandwidth by over 98% compared to deep federated baselines.

***Index Terms*—Federated Learning, Intrusion Detection Systems (IDS), Internet of Things (IoT), MQTT Protocol, Differential Privacy, Secure Aggregation, Tabular Data Synthesis, Probabilistic Graphical Models.**

I. Introduction

A. Background and Motivation

The Internet of Things (IoT) connects physical objects, sensors, and various devices through communication networks, enabling data-driven services in domains such as healthcare, industrial automation, and smart environments. Alongside the expansion of IoT deployments, the number of connected components exposed to cybersecurity threats increases correspondingly [1]. The Message Queuing Telemetry Transport (MQTT) protocol is commonly used for lightweight publish-subscribe communication in IoT environments. However, MQTT-based systems can become targets for denial-of-service (DoS) and distributed denial-of-service (DDoS) attacks. The DoS/DDoS-MQTT-IoT dataset, collected from a physical IoT testbed, provides normal MQTT traffic alongside ten DoS attack scenarios to facilitate the study of attacks in this context [2]. Intrusion Detection Systems (IDS) monitor network activity to identify traffic or behavior exhibiting malicious signs. Machine learning and deep learning methods have been utilized and researched for this task due to their ability to learn characteristic patterns directly from network traffic. Their effectiveness depends partly on the availability and representativeness of the training data [3]. Federated Learning (FL) allows multiple clients to collaboratively train a shared model while keeping their data local; a central server distributes the global model and aggregates updates from clients across training rounds [4]. This approach is highly suitable when IoT data is distributed across multiple devices or locations, although communication and computational costs remain critical factors to consider in resource-constrained environments [5], [6]. Keeping data local also does not equate to completely eliminating privacy concerns arising from the exchange of model updates [7].

Privacy-preserving synthetic data generation is a potential solution to address the shortage of training data for rare attack classes in IoT networks while still ensuring strict security guarantees. Rather than relying on deep neural networks that require substantial computational resources and face convergence difficulties on edge devices [8], the probabilistic graphical model paradigm (Probabilistic Graphical Models) approaches the problem by decomposing tabular data into low-dimensional marginal frequency distribution slices (low-dimensional marginals) [9]. Through the Chow-Liu dependency tree structure (Chow-Liu Tree) [10], this approach effectively captures the core correlations between traffic features and attack labels without being affected by the curse of dimensionality. When combined with Differential Privacy (DP) [9] on contingency tables and consistent graphical inference mechanisms (such as Private-PGM), the system can reconstruct a valid global distribution to generate conditional traffic data, establishing a

foundation to enhance the performance of downstream intrusion detection systems (IDS) [11], [12].

B. Problem Statement

Building an effective IDS for IoT networks faces multiple challenges related to data privacy, the representation of attack classes, and the practical costs of the model training process. These challenges are particularly evident when training data is distributed among clients and synthetic attack samples are considered as a supplementary detection solution. The first problem relates to privacy risks during federated training. Although FL avoids the periodic transfer of raw local data, clients must still exchange model updates with the server. Previous studies have demonstrated that shared gradients can leak information about training data in certain attack scenarios [7]. Therefore, sharing model weights instead of raw data cannot be considered a comprehensive privacy protection measure. Depending on the threat model and the type of information to be protected, supplementary mechanisms may be mandatory. The second problem concerns the representation of attack classes in the training data. Class imbalance can bias the learning process towards majority classes, making it difficult to identify classes that account for a low percentage [3]. For the DoS/DDoS detection problem in MQTT, the impact of this issue depends on the dataset version, label mapping, preprocessing pipeline, and data partitioning strategy for clients. Therefore, the class distribution of the dataset after selection and preprocessing must be empirically verified rather than assumed by default. If some attack classes lack data, synthetic samples can supplement training examples. However, merely increasing the number of samples does not prove that the generated data is useful; their actual contribution must be evaluated through data quality metrics and the detection performance of the downstream IDS model.

The third challenge pertains to the hardware constraints of edge devices, the "curse of dimensionality," and the intricate trade-off among privacy protection, data quality, and operational overhead: on the one hand, deploying traditional distributed deep generative models (such as GANs or CVAEs) requires substantial computational resources (GPU/CPU) and large RAM capacity at client nodes, which is infeasible for hardware-constrained edge IoT devices [8]. On the other hand, tabular MQTT network traffic inherently possesses a high-dimensional structure with numerous interdependent attributes. Applying Differential Privacy (DP) directly to this high-dimensional attribute space readily leads to a state-space explosion—reaching up to 10^{19} cells [11]—causing the injected differential privacy noise to completely overwhelm the original data signal and severely degrade the fidelity of the generated distributions [9].

A direct consequence is a severe tug-of-war among three factors: the degree of privacy protection (privacy budget ε), synthetic data utility for downstream IDS models, and computational/operational overhead across the IoT infrastructure. Specifically, tightening privacy guarantees (reducing the budget ε) injects heavy DP noise that destroys the core feature correlations of rare DoS/DDoS attack classes [9], [12]. Conversely, increasing the privacy budget and expanding model capacity to improve the quality of generated data causes computational costs and system runtime to surge drastically [12]. Consequently, finding a tabular data generation solution that simultaneously ensures rigorous privacy guarantees, remains computationally lightweight for edge devices, and preserves critical feature characteristics for rare attack classes remains a major, unresolved technical bottleneck.

C. Research Objectives and Questions

1) Objectives (O1–O4): This study proposes PP-FedData—a privacy-preserving federated data generation framework for intrusion detection systems (IDS) in IoT-MQTT networks. The research aims to achieve four primary objectives (O1–O4):

- **Objective O1 – Federated Architecture Construction:** Model each IoT Gateway device as a client within the Federated Learning network (FL Client). Each gateway stores raw packet traffic data locally and only transmits protected statistics to the Aggregation Server under an honest-but-curious threat model.
- **Objective O2 – Marginal-based Federated Generator Development (FedDP-Marginal):** Extend marginal-based privacy-preserving data synthesis methods (such as MST and PrivBayes) to the federated learning environment: each gateway releases noisy per-class count tables across three communication rounds. Noise can be partitioned among gateways and accumulated via the Secure Aggregation mechanism (SecAgg$^+$) (corresponding to Distributed Differential Privacy using discrete Skellam noise), or fully added locally at each gateway (Local DP). Synthetic data records are subsequently sampled from a Chow-Liu dependency tree structure trained on the released marginal statistics tables.
- **Objective O3 – Deep Generative Baseline Implementation:** Implement a federated Conditional Variational Autoencoder (CVAE) model as a deep generative baseline. The CVAE model is trained using the FedAvg algorithm and protected by DP-SGD, SecAgg$^+$, or a combination of both, under identical data partitioning and evaluation protocols to enable a head-to-head empirical comparison against the FedDP-Marginal generator.
- **Objective O4 – Comprehensive Evaluation and Deployment Guidance Formulation:** Comprehensively evaluate all privacy-preserving configurations on a held-out real test set across five dimensions (utility, fidelity, privacy, overhead, and statistical rigor). From these empirical findings, establish a transparent decision rule to map practical deployment constraints (server/client trust assumptions, privacy budget ε, communication bandwidth, prioritized attack classes) into their corresponding optimal configurations.

2) Research Questions (RQ1–RQ4): To fulfill the four aforementioned objectives, this research focuses on addressing four core research questions:

- **RQ1:** When raw network traffic data cannot be centralized, how much IDS utility does the synthetic data generated

by the federated generator retain compared to an IDS trained on centralized real data? Does augmenting training sets with synthetic data improve detection performance when real data is already available (either centrally or at individual gateways)?
- **RQ2:** Under the same privacy budget (ε, δ), how does the federated marginal-based generator (FedDP-Marginal) compare to the federated CVAE model trained with DP-SGD regarding downstream utility, detection capabilities on rare attack classes, and network communication overhead?
- **RQ3:** What overhead does each protection mechanism incur under DP, SecAgg$^+$, or their combination, and at what granularity does it provide privacy protection? How do the unit of privacy protection (packet, TCP stream, or capture session) and the number of honest gateways alter privacy guarantees and operational costs?
- **RQ4:** Which protection configuration (combining DP, SecAgg$^+$, noise distribution, and privacy protection unit) should be selected for specific real-world deployment requirements?

D. Significance and Contributions

This study addresses two fundamental bottlenecks in collaborative intrusion detection across IoT-MQTT networks: (1) rare attack classes are sparsely and non-identically distributed across IoT Gateway devices (Non-IID), and (2) gateways cannot share raw packet traffic data due to privacy concerns.

The study offers three primary scientific and practical contributions:

- **Marginal-Based Distributed Generator (FedDP-Marginal):** Proposes a distributed synthetic data generation solution in which marginal statistics released by individual gateways are simultaneously protected by formal Differential Privacy (DP) and Secure Aggregation (SecAgg$^+$).
- **Controlled Benchmark:** Conducts a direct, head-to-head empirical comparison between the marginal-based generator (FedDP-Marginal) and the deep generative baseline (Federated CVAE) under identical data partitioning schemes, consistent privacy-preserving mechanisms, and a rigorous evaluation protocol.
- **Requirement-Driven Configuration Guide:** Proposes a transparent decision guideline to assist IoT system administrators in selecting the most suitable privacy configuration based on their specific threat model, mutual trust assumptions, and practical resource budgets.

Explicitly Stated Limitations and Privacy Guarantees: The privacy guarantees in this study are precisely and transparently defined:

- **Differential Privacy (DP)** provides mathematical upper bounds limiting the amount of information that released statistical tables or models can leak regarding a single data record (by default at the packet level, or optionally at the TCP stream or capture session level under specific assumptions). The DP mechanism does not anonymize class labels and does not enforce privacy guarantees on the hyperparameter tuning process conducted on validation data.
- **Secure Aggregation (SecAgg$^+$)** only guarantees the confidentiality of individual gateway updates/statistics with respect to the Aggregation Server; it does not prevent what the final model or generated synthetic data might leak about the aggregate dataset.

E. Scope and Limitations

This research is scoped within the following specific experimental and technical boundaries:

- **Dataset and Data Granularity:** Utilizes the DoS/DDoS-MQTT-IoT dataset at the packet level (where each record corresponds to a single network packet and includes six traffic class labels: one Normal class and five DoS/DDoS attack families).
- **Federated Simulation Architecture:** A network of five simulated IoT Gateway devices is deployed under an uneven data distribution across nodes (Dirichlet non-IID partitioning).
- **Generative Models and Privacy Mechanisms:** Evaluates and compares two generators: the primary marginal-based federated generator (FedDP-Marginal) and the deep generative baseline (Federated CVAE). Enforces Differential Privacy (DP) across privacy budget levels $\varepsilon \in \{1, 5, 10\}$ with $\delta = 10^{-5}$, Secure Aggregation (SecAgg$^+$), as well as a hybrid configuration combining both mechanisms.
- **Experimental Infrastructure:** The entire federated training workflow is simulated on a single physical host machine using the Flower framework.

Despite achieving rigorous empirical findings, this study is subject to several scientific limitations:

- **Single-Dataset Reliance:** All empirical evaluation results and configuration selection guidelines are derived exclusively from the DoS/DDoS-MQTT-IoT dataset; therefore, generalizability to other IoT application protocols (such as CoAP or HTTP/2) requires further empirical validation.
- **Packet-Level Independence Assumption:** Packets are modeled independently as tabular records, which does not fully capture the temporal/sequential dependencies inherent in an end-to-end communication session or stream.
- **Default Record-Level Privacy Scope:** The Differential Privacy (DP) guarantee applies by default at the individual record level (a single packet). Extending protection to an entire TCP stream or a full packet capture session represents an advanced configuration option that necessitates a trade-off resulting in degraded synthetic data utility.
- **Privacy Budget ε Exclusions:** The privacy budget ε covers only the released count statistics and trained models. Auxiliary metadata such as attack class labels, global preprocessing statistics, and validation-based hyperparameter tuning fall outside the theoretical guarantee of ε.

- **Simulation Environment:** Conducting the simulation on a centralized physical machine does not fully capture real-world network latency, packet loss rates, or the physical hardware constraints of resource-limited edge IoT devices.
- **Constrained Threat Model:** The study operates under an honest-but-curious threat model across all gateways. Sophisticated adversarial threat scenarios such as data/model poisoning attacks or Byzantine clients are outside the scope of this investigation.

F. Paper Organization

The remainder of this paper is structured as follows: Section II reviews related work and identifies the research gaps. Section III presents the system model and the proposed generative defense methodology. Section IV details the experimental evaluation and empirical results. Section V discusses findings and trade-offs, and Section VI concludes the paper with future research directions.

II. Related Work and Research Gaps

A. Privacy-Preserving Mechanisms in Federated Learning

1) Landscape and Scope of Research: The integration of Federated Learning (FL) into Intrusion Detection Systems (IDS) for IoT environments has resolved raw data centralization, yet the model weight sharing process itself still harbors data leakage risks [13]. Current studies classify Privacy-Preserving Federated Learning (PPFL) techniques into five main categories: Differential Privacy (DP), Homomorphic Encryption (HE), Secure Multi-Party Computation (SMPC), Trusted Execution Environments (TEE), and Blockchain [13]. Although TEEs provide a secure execution environment and Blockchain ensures transparency, they require specialized hardware or complex consensus mechanisms unsuitable for the constrained resources of edge IoT devices [14]. Therefore, this study implements DP and SMPC-based secure aggregation (SecAgg$^+$); HE is reviewed as an alternative but not implemented due to its prohibitive computational cost on constrained devices [15].

2) Trusted Execution Environments (TEE): The Trusted Execution Environment (TEE) provides an isolated and secure hardware computing enclave for processing sensitive telemetry, effectively preserving confidentiality. However, this technology depends heavily on specialized hardware security modules. This strict hardware requirement makes TEE cumbersome and unsuitable for ubiquitous deployment across heterogeneous IoT networks comprising massively resource-constrained devices [13].

3) Blockchain-Based Mechanisms: Blockchain preserves data privacy through decentralized ledgers and cryptographic consensus that ensure tamper-resistance and eliminate intermediaries. However, these architectures suffer from fatal performance weaknesses due to low throughput, high latency, and expensive verification costs stemming from continuously growing immutable storage. These latency and storage overheads make blockchain impractical for lightweight, multi-device IoT environments that demand rapid detection response times [13].

4) Homomorphic Encryption (HE): HE enables arithmetic operations directly on encrypted ciphertexts, ensuring model updates from clients remain confidential prior to server aggregation without decryption [15]. However, the computational complexity and communication bandwidth expansion of HE algorithms are exceptionally heavy, drastically increasing system latency [15]. This renders HE unfeasible for edge IDS requiring real-time response under tight resource constraints [15].

5) Differential Privacy (DP): Unlike the heavy overhead of HE, Differential Privacy (DP) offers high scalability. The core principle of DP is to mask the contribution of any individual record by injecting calibrated random perturbation into local updates before transmission [16]. DP is exceptionally lightweight, incurs negligible transmission overhead, and scales easily to large client cohorts [16]. The trade-off is that noise injection degrades model parameter precision, necessitating careful calibration of the privacy budget ε [16].

6) Secure Multi-Party Computation (SMPC/SecAgg): Conversely, Secure Aggregation (SecAgg) protocols based on SMPC resolve the accuracy degradation of DP. SMPC prevents the server from inspecting individual client updates by ensuring that only the aggregated sum is revealed [17]. SecAgg introduces zero statistical perturbation; under the fixed-point quantization utilized in SecAgg$^+$, the aggregate differs from the plain sum only by bounded quantization error [18]. Although the original SecAgg protocol suffered from multi-round interaction bottlenecks scaling quadratically with client counts [19], the SecAgg$^+$ protocol resolves these issues by replacing complete communication graphs with k-regular graphs, achieving polylogarithmic communication and computational complexity [18].

7) Hybrid Privacy-Preserving Systems: Gutti et al. [20] (combining DP and SecAgg via the HADA algorithm) demonstrated that pairing DP (to prevent post-aggregation inference) with SecAgg (to secure transmission channels) is a promising paradigm. However, stacking these security layers introduces substantial hyperparameter tuning and coordination complexity [20]. Evaluating independent and joint performance trade-offs under realistic edge conditions is therefore essential before deploying hybrid pipelines.

8) Distributed Differential Privacy: Distributed DP combines the strengths of both paradigms: each client adds only a fraction of the required noise, and secure aggregation reveals only the collective sum containing the full calibrated perturbation [21], [22]. Because secure aggregation operates over modular integer lattices, discrete noise distributions are preferred; the sum of independent Skellam variables remains Skellam, permitting exact privacy accounting after modular aggregation [22]. Distributed DP is particularly well-suited for single-sum releases such as contingency count tables.

B. Generative Data Augmentation for Intrusion Detection

1) Taxonomy and Core Concepts: Generative AI paradigms in IoT-IDS learn the underlying multivariate probability distributions of network telemetry to automatically synthesize realistic

traffic samples whose structural and statistical attributes align with genuine traffic [23].

Core problems addressed by generative augmentation:

- **Mitigating Extreme Class Imbalance:** Real-world IoT traffic exhibits severe class skewness, where over 90–99% of traffic comprises benign activities, while rare DoS/DDoS attacks constitute negligible fractions. This deficit biases downstream classifiers toward majority classes, generating unacceptable false negatives on stealthy threats [23].
- **Surpassing Interpolation-Based Oversampling:** Traditional oversampling algorithms such as SMOTE [24] perform linear interpolation within localized convex hulls. Deep generative models capture global distributions and explore under-represented regions of feature space to synthesize zero-day attack variants. Furthermore, SMOTE requires centralized access to neighboring minority instances, making it inapplicable to federated environments where individual edge gateways possess few or no attack records [24].
- **Data Augmentation and Restructuring:** Systematic surveys show that up to 89% of studies leverage generative models for data augmentation and class rebalancing, improving classification generalization without centralizing raw traffic [23].

Primary taxonomies of generative architectures in IDS:

- **Generative Adversarial Networks (GANs):** Structured around an adversarial minimax game between a Generator and a Discriminator. Modeling sharp distributions makes GANs the predominant architecture (58%) in generative IoT-IDS [23].
- **Variational Autoencoders (VAEs):** Encoders map high-dimensional traffic into continuous latent manifolds, while Decoders reconstruct inputs. VAEs enforce structured prior distributions, permitting stable sampling without minimax instabilities [23].
- **Transformer-Based Models:** Multi-head self-attention mechanisms model complex sequential dependencies across packet streams [23].

2) Critical Analysis of Generative Approaches: Existing literature reveals noticeable bottlenecks when deploying generative models in decentralized edge settings [23]:

- **Tabular Diffusion (Tab-DDPM):** Manzano et al. [14] integrated tabular diffusion on the BoT-IoT benchmark. While Tab-DDPM achieves high fidelity, iterative denoising over hundreds of reverse steps incurs 32 seconds per sample on edge nodes (1 CPU core, 2 GB RAM), which is infeasible for line-rate inspection.
- **Policy-Guided GANs:** Yang et al. [25] employed RL-guided GANs for DDoS augmentation. However, the multi-component architecture requires 2.726 seconds per batch and 19.6 GB of GPU memory on KDDcup99, without isolating standalone edge inference latency.
- **Adversarial Instability:** Deng et al. [23] observed that GAN training is prone to mode collapse, wasting node resources by repeatedly generating duplicate records.
- **Resampling Baselines (Bootstrapping):** Jacob and Sultana [26] used bootstrapping to balance BoT-IoT and CSE-CIC-IDS2018 datasets. While lightweight, duplication does not expand feature diversity, increasing overfitting risks.
- **Centralized Cloud VAE-GAN:** Anandhi and Srikanth [27] synthesized 51,200 Low-Rate Stealth Attack samples in the cloud, raising recall from 62% to 98%. However, cloud synthesis violates federated privacy principles, and compressing models below 6 bits degraded detection sensitivity.
- **Lightweight Edge GANs (TinyGAN):** Durga et al. [28] deployed TinyGAN for IoMT malware detection. However, all classes below 0.5% prevalence were discarded, avoiding the fundamental minority imbalance challenge.

C. Privacy-Preserving Generative Defenses and Identified Research Gaps

1) Privacy-Utility Dilemma in Edge Generative IDS: Federated IDS on edge networks must protect client telemetry while coping with severe minority-class imbalance and non-IID traffic distributions. Differential privacy and cryptographic protocols bound gradient inversion and membership inference risks [29] but do not correct class skew. Augmenting under-represented attack classes with federated generative models mitigates this imbalance, yet generative architectures may memorize training records and expose sensitive network traces through parameter exchanges or synthesized flows [29]. Edge defenses must therefore limit the utility loss of synthetic data while providing formal client-side privacy guarantees.

Cryptographic mechanisms, such as secure multi-party computation (MPC) and threshold secret sharing, preserve numerical precision without perturbation error [29]. However, secure aggregation of high-dimensional neural parameters requires iterative multi-round communication, which burdens resource-constrained IoT edge devices [29].

2) State-of-the-Art in Generative PPFL-IDS: A literature search of IEEE Xplore and SpringerLink (2022–2026) for studies combining federated learning, intrusion detection, and generative models (GAN or VAE) identified five empirical implementations, all from IEEE [30]–[34]. None applies cryptographic secure aggregation to uplink generator updates. Salim et al. [32] (global VAE aggregation), Jiao et al. [33] (local CGAN gradients), and Shettar et al. [34] (central aggregator with moments-accountant clipping at $\varepsilon = 4.8$ and $\delta = 10^{-5}$) rely on DP alone. He et al. [31] combine local gradient noise with blockchain Proof-of-Authority consensus across edge servers, which ensures aggregation integrity and auditability rather than parameter confidentiality.

Yao et al. [30] use a hybrid design: proactive dynamic (t, n)-threshold secret sharing with $t = 4$ and $n = 5$ protects only downlink synthetic broadcasts, whereas uplink client gradients rely on Gaussian DP. They report execution measurements for key distribution and sample reconstruction (Table VI), but secret sharing covers downlink tabular samples, not full model-

parameter aggregation. Outside the primary search, Hamouda et al. [35] likewise omit secure aggregation and identify cryptographic model protection as an open challenge. Because none of these works benchmarks cryptographic aggregation against DP on model weights, their use of DP reflects a design choice rather than a measured trade-off.

The effect of DP perturbation on downstream detection depends on the privacy budget ε, gradient clipping threshold C, and minority-class sample density, yet existing studies do not isolate these factors. Utility can remain high at a moderate budget or with large synthetic volumes: Shettar et al. [34] report a 96.8% detection F1-score under central DP ($\varepsilon = 4.8$), and Hamouda et al. [35] observe a 0.25-percentage-point accuracy reduction at $\varepsilon = 1.0$ when applying DP-SGD to the downstream global classifier with 50,000 synthetic samples per class. Generative augmentation can also raise rare-class detection from sparse baselines: He et al. [31] report that CGAN augmentation improved Infiltration F1-score from 33.33% to as high as 98.01%, starting from 29 baseline training samples. However, Hamouda et al. [35] applied DP to the classifier rather than the generator, and neither He et al. [31] nor Jiao et al. [33] report formal numerical (ε, δ) budgets. Because generator-side DP must spread privacy noise across iterative training steps, how strict bounds ($\varepsilon \leq 1.0$) interact with extreme class scarcity in federated generative architectures remains an open empirical question.

3) Marginal-Based Tabular Synthesis vs. Deep Models: Marginal-based tabular synthesis offers an alternative to deep generative models. Instead of optimizing non-convex neural network weights by backpropagation, marginal synthesizers select and measure low-order contingency tables and reconstruct the data distribution with probabilistic graphical models. Beyond network IDS research, a cross-domain search of the ACM Digital Library, PMLR, and arXiv identified distributed extensions: Maddock et al. [36] introduce FLAIM, which adapts the Adaptive and Iterative Mechanism (AIM) to distributed clients under zero-concentrated differential privacy (ρ-zCDP) and compares DP with secure multi-party computation (DistAIM); Pentyala et al. [37] present CaPS, a "DP-in-MPC" framework in which secret sharing protects marginal selection and measurement; and Li et al. [38] propose HeteroFedSyn, which compresses two-way marginal dependencies by random projection under distributed perturbation.

Marginal synthesizers avoid the adversarial minimax training instability of GANs but carry their own privacy-utility trade-offs. Iterative selection-and-measurement algorithms (e.g., AIM and Private-PGM) divide the total privacy budget ε across rounds, and distributing measurements across federated clients compounds perturbation noise [36], [37]. In sparse discrete categories with small contingency cell counts, additive Laplace or Gaussian noise yields a low signal-to-noise ratio. Restricting measurements to low-order marginals (typically two-way dependencies [38]) truncates higher-order multivariate correlations, and continuous flow variables require discretization, which introduces binning artifacts.

Federated marginal generators have so far been evaluated mainly on general-purpose tabular benchmarks from demographic, financial, and biomedical domains (e.g., Adult, Census, Credit Card Fraud, COMPAS, Diabetes) [36]–[38]. FLAIM [36] also used the legacy DARPA Intrusion Detection benchmark but evaluated only workload error and negative log-likelihood, not downstream minority-class intrusion classification. Decentralized tabular GANs with cryptographic primitives also exist: Duan et al. [39] combine Paillier additively homomorphic encryption for global multimodal distribution encoding with DP during GAN training, but evaluate statistical utility and membership inference resistance on legacy tabular data (KDD Cup 99), not modern protocol telemetry.

4) IoT-MQTT Protocol Telemetry and Hypotheses: IoT-MQTT telemetry combines structured, low-cardinality discrete protocol fields (e.g., message types, QoS levels, retain flags) with continuous flow metrics (e.g., payload lengths, burst durations). Sparse discrete feature combinations form low-count contingency cells, where both generative families risk degradation under DP noise, making this a suitable setting for examining these trade-offs. We hypothesize that, under generator-side DP at strict budgets ($\varepsilon \leq 1.0$), a federated marginal synthesizer achieves higher minority-class detection F1-score than a federated deep generative model for attack classes defined by low-order (two-way) discrete header correlations (e.g., MQTT Unauthorized Connect, Malformed Header). We expect the reverse for volumetric or timing-based attacks (e.g., MQTT Publish Floods) governed by multi-attribute continuous flow dynamics. We test this across five random seeds, equalizing privacy budgets by converting all mechanisms to (ε, δ)-DP under Rényi differential privacy accounting, and compare class-specific F1-scores with a Wilcoxon signed-rank test (treating a margin $|\Delta\mathrm{F1}| \geq 2.0$ percentage points at $p < 0.05$ as statistically significant).

5) Identified Research Gaps: To the best of our knowledge, the systematic survey indicates three distinct research gaps:

- **Gap 1 – Marginal Synthesis for Protocol Telemetry:** Deep generative models have been evaluated on MQTT-bearing datasets (e.g., Hamouda et al. [35], federated conditional GAN on Edge-IIoTset), but no prior study evaluates a federated marginal-based generator on modern IoT-MQTT telemetry or examines how mixed discrete-continuous features affect marginal synthesis relative to deep models.
- **Gap 2 – Controlled Architectural Comparison:** No empirical study compares a federated marginal-based synthesizer with a federated deep generative model (e.g., CGAN, VAE) under identical non-IID client partitions, minority-attack imbalance ratios, and calibrated generator-side privacy budgets (ε, δ).
- **Gap 3 – Downstream Detection Trade-offs:** Prior marginal studies compared DP with cryptographic aggregation on workload error, negative log-likelihood, runtime, and communication bandwidth [36], [37], and decentralized GANs were evaluated for statistical utility and membership inference resistance [39]. No study quantifies how choosing between DP and cryptographic secure

aggregation affects downstream minority-class intrusion detection under matched federated edge conditions.

D. Summary and Positioning of This Work

This study directly resolves the aforementioned gaps by presenting PP-FedData. We evaluate communication overhead, synthetic sample fidelity, and downstream minority-class intrusion detection for both marginal and deep generative models under differential privacy and cryptographic secure aggregation on realistic edge IoT networks.

III. Research Design and Methodology

A. Research Type and Experimental Control

This study is a quantitative, controlled experimental study carried out on one published network-traffic dataset, DoS/DDoS-MQTT-IoT [2]. In the terminology of experimental research methodology, it combines controlled experiments with archival research: the independent variables (generative architecture, privacy protection mechanism, and differential privacy budget) are systematically manipulated, while the underlying network traffic comprises recorded real-world traces rather than synthetic traffic generated de novo for this study.

1) Research Object: The object of study is a distributed privacy-preserving data synthesis framework, PP-FedData, which enables multiple IoT/MQTT gateways to collaboratively construct synthetic network telemetry without releasing raw packets. Consequently, downstream Intrusion Detection Systems (IDS) can be trained on representative traffic distributions that no single edge gateway could pool independently. PP-FedData is evaluated across four primary dimensions:

1) **Downstream Utility:** The detection accuracy of the synthetic data for downstream classifiers;
2) **Formal Privacy Guarantees:** The mathematical bounds and empirical leakage resistance of released artefacts;
3) **System Overhead:** Communication bandwidth (uplink/downlink) and computational latency per client;
4) **Operational Fitness:** The compliance of each configuration with practical edge deployment requirements.

2) Experimental Control: Internal validity is maintained by holding all factors invariant except the specific generative route and privacy mechanism under study. Every configuration in the experiment matrix (Section III-D) is evaluated under identical conditions:

- **Identical Data Partitioning:** A locked train/validation/test split partitioned strictly by capture group. The real test split is locked and never accessed during hyperparameter tuning (train: 88,500 rows; validation: 12,000 rows; test: 30,000 rows; class quotas strictly enforced);
- **Consistent Non-IID Skew:** Partitioning of training rows over $K = 5$ edge clients drawn from a Dirichlet distribution with concentration parameter $\alpha = 0.5$ [40], fixed deterministically by random seed;
- **Uniform Preprocessing:** A standardized pipeline ($\log(1+x)$ transformation of skewed numeric attributes, standard normal scaling, clipping at $\pm 5\sigma$, and one-hot encoding of categorical headers), producing exactly 60 input feature dimensions;
- **Standardized Downstream Classifiers:** A Random Forest [41] with 200 estimators and a Multilayer Perceptron (MLP) with hidden layers $[128, 64]$;
- **Multi-Seed Replication:** Identical seed triad $\{0, 1, 2\}$ governing weight initialization, client partitioning, and synthetic sampling;
- **Fixed Real Test Set:** All reported utility metrics are evaluated on the exact same held-out real test set.

Under this design, differences in downstream performance are directly attributable to the generative model and privacy mechanism. Extraneous sources of variance, such as hyperparameter optimization and capture group boundaries, are addressed in Section III-N.

3) Research Questions and Design Mapping: Table II maps each research question to its comparative configurations, evaluation protocols, and primary measures. Evaluation protocols follow standardized notation: TRTR (Train on Real, Test on Real), TSTR (Train on Synthetic, Test on Real), TAug (Train on Real augmented with Synthetic), TAugR (Augmentation restricted to rare minority classes), and local augmentation (supplementing a client's private telemetry with synthetic samples for unobserved classes).

The rare attack classes comprise `DELAYED`, `SYN`, `INVALID`, and `WILL`, representing the four MQTT attack families whose training quota is at most 3,000 records. Macro-F1 is computed as the unweighted mean of per-class F1-scores across all six classes, while binary F1 evaluates discrimination between `NORMAL` traffic and the aggregated attack classes.

4) Overall Research Workflow: The experimental workflow proceeds through five sequential phases (Fig. 1): data preparation with integrity leakage checks; non-IID federated partitioning; dual-route generative training; downstream evaluation across multiple protocols; and multi-criteria aggregation into a Pareto scorecard.

5) Evolution of the Design: The study was originally conceptualized around a federated conditional variational autoencoder (CVAE) [42]. Following initial empirical evaluation, the framework transitioned to a marginal-based generator, *FedDP-Marginal*, as its primary generative mechanism, while retaining the federated CVAE as a baseline. This architectural pivot is formally documented in the project deviation log (`docs/SPEC_DEVIATIONS.md`, entries O1.9 and O2.3). Maintaining the CVAE configurations in the benchmark matrix ensures a rigorous, head-to-head comparison under identical federation topologies, differential privacy budgets, and secure aggregation environments.

B. System Architecture and Generator Routes

1) System Roles: As illustrated in Fig. 2, the PP-FedData architecture comprises three primary entities:

1) **Edge Gateways ($K = 5$ FL Clients):** Each gateway observes local MQTT traffic and maintains exclusive custody of its telemetry. Due to non-IID Dirichlet partitioning, individual gateways frequently exhibit severe

TABLE I
TAXONOMY AND COMPARISON OF EXISTING GENERATIVE PRIVACY-PRESERVING INTRUSION DETECTION FRAMEWORKS

Study	Generative Architecture	Privacy Primitive	Target Domain / Dataset	Aggregation Mechanism	SecAgg vs. DP Benchmark
Yao et al. [30]	Improved GAN	Gaussian DP + (t, n)-TSS	IoT Edge Networks	DP Uplink, TSS Downlink	No
He et al. [31]	Conditional GAN	Local DP + PoA Blockchain	UAV Networks	Blockchain Consensus	No
Salim et al. [32]	Variational Autoencoder	Central DP	Satellite Communications	Global VAE Average	No
Jiao et al. [33]	Conditional GAN	Local DP Gradients	Industrial IoT (Drones)	Federated Averaging	No
Shettar et al. [34]	SYNAPSE Pipeline	Central Moments DP ($\varepsilon = 4.8$)	Edge Security Telemetry	Central Aggregator	No
Hamouda et al. [35]	Federated CGAN	Classifier DP-SGD	Industrial IoT (Edge-IIoTset)	Plain FedAvg	No
Maddock et al. [36]	FLAIM (Marginal/AIM)	ρ-zCDP vs. MPC	DARPA / Tabular Benchmarks	DistAIM / SMPC	Workload Error Only
Pentyala et al. [37]	CaPS (DP-in-MPC)	Secret Sharing + DP	Demographic / Medical Data	SMPC Marginal Select	Workload Error Only
Li et al. [38]	HeteroFedSyn	Distributed Perturbation	General Tabular Data	Random Projection Sum	No
Duan et al. [39]	HT-Fed-GAN	Paillier HE + DP	KDD Cup 99	Homomorphic Sum	Statistical Utility Only
This Work	**FedDP-Marginal vs. CVAE**	**DP + SecAgg$^+$ (Skellam)**	**IoT-MQTT Telemetry**	**Cryptographic vs. DP**	**Yes (Downstream F1)**

TABLE II
RESEARCH QUESTIONS AND EXPERIMENTAL DESIGN MAPPING

RQ	Core Question	Compared Configurations	Protocols	Primary Measures
RQ1	Does federated synthetic data improve IDS performance relative to limited local real data?	B0, B1a, B1b (pooled real references) vs. B3, M2, MG (synthetic routes)	TRTR, TSTR, TAug, TAugR, local augmentation	Macro-F1 across 6 classes; mean recall on rare minority attack classes.
RQ2	How do marginal-based and deep generative routes compare under identical DP budgets?	M1, M1o, M3, M3o (federated CVAE with DP) vs. MG, at $\varepsilon \in \{1, 5, 10\}$	TSTR	Macro-F1; rare-class recall; communication volume (MB/round); convergence rounds.
RQ3	What are the quantitative utility and resource trade-offs incurred by privacy mechanisms?	B3 vs. M1 (cost of DP); B3 vs. M2 (cost of SecAgg); MGs vs. MGl; suffixes `-strm`, `-cap`, `-t`	TSTR, privacy audit, overhead monitoring	Privacy degradation as a function of honest clients h; membership inference AUC; transmitted MB; runtime.
RQ4	Which configuration optimally satisfies a given multi-objective deployment requirement?	All verified configurations in the scorecard	Requirement-driven rule (Section III-E)	Compliance with operational requirement vector $\mathcal{R}$.

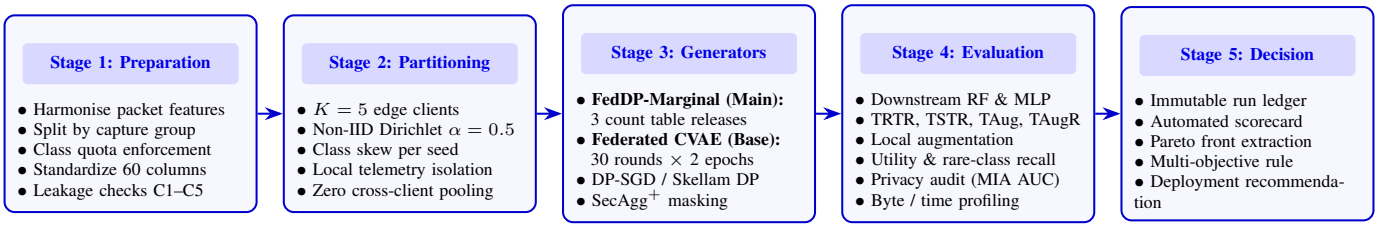


Fig. 1. Overall five-stage experimental workflow of the PP-FedData framework.

data sparsity or complete absence of specific rare attack classes.

2) **Aggregation Server:** The central coordinator orchestrates communication rounds, computes secure aggregates of client submissions, reconstructs global generative models, and generates synthetic records. The server is modelled as *honest-but-curious* (Section III-C).
3) **Downstream IDS:** Classifiers (Random Forest and MLP) are trained on generated synthetic traffic (TSTR) or hybrid datasets (TAug, TAugR, local augmentation) and evaluated on the locked real test set.

2) Synthesis Locality and Server Observability: Synthetic data records are synthesized centrally at the aggregation server, conditional on specified class labels, either by decoding latent vectors via the global CVAE decoder or by ancestral sampling over the reconstructed global dependency tree and marginal tables. Edge gateways do not generate synthetic traffic locally.

The visibility of client updates to the aggregation server is strictly determined by the active privacy configuration:

- **SecAgg$^+$ Protocols (M2, M3, MGs, MGr):** Utilizing Flower's SecAgg$^+$ [18], [43], the server observes only masked parameter vectors, reconstructing strictly the additive sum while learning zero information regarding individual client contributions;
- **Plain FedAvg (B3, M1):** The server observes raw individual client model updates. In M1, updates are perturbed locally via DP-SGD;
- **Local DP (MGl):** Gateways inject full calibrated noise into local tables prior to plaintext transmission, allowing the server to observe individual noisy tables without cryptographic masking;
- **Simulated Distributed DP (MGd, MGb):** The distributed

summation is computed within a single execution process to benchmark accounting efficiency.

3) Comparison of the Two Generator Routes: Table III contrasts the fundamental architectural mechanisms of the two routes. FedDP-Marginal exchanges discrete frequency counts across three synchronized releases, whereas the federated CVAE exchanges continuous parameter weight vectors over 30 iterative FedAvg rounds.

This operational divergence is decisive for differential privacy accounting. In FedDP-Marginal, each release represents a linear additive summation with bounded global sensitivity; calibrated noise can therefore be divided among clients and reconstituted exactly via secure aggregation. In contrast, DP-SGD [44] introduces non-linear gradient transformations across successive local optimization epochs, where noise added by one client interacts non-linearly with others; distributed DP is consequently mathematically invalid for the CVAE route (`SPEC_DEVIATIONS.md`, entry O2.1).

For FedDP-Marginal, secure summation is implemented via a lightweight client-side patch: stock Flower SecAgg$^+$ transmits masked vectors as 64-bit integers. PP-FedData casts masked vectors to unsigned 32-bit integers (`uint32`) following modular reduction, leveraging CPU register wrap-around to execute exact modulo 2^{32} addition without altering installed external packages (`SPEC_DEVIATIONS.md`, entry O2.2).

C. Threat Model, Trust Assumptions, and Privacy Units

1) Adversary and Trust Assumptions: The aggregation server is modeled as *honest-but-curious* (semi-honest) [46]: it faithfully executes protocol steps but inspects all incoming payloads to infer private telemetry. Gateways maintain physical control of their own data. We analyze four trust settings:

1) **Base Setting:** All K gateways adhere to protocol and inject their assigned noise share ($1/K$ of the total required variance in distributed variants);
2) h **Honest Gateways:** The privacy guarantee is audited under scenarios where only $h \in \{1, \ldots, K\}$ clients inject noise, while the remaining $K-h$ clients collude with the server or drop out. Effective ε increases monotonically as h decreases;
3) **Threshold** t**:** Noise is calibrated such that target (ε, δ) holds whenever at least t gateways contribute, with each client injecting $1/t$ of total variance (default $t = K$);
4) **Local DP (MGl):** Each client adds the full required noise independently [47], achieving protection without trusting peer gateways.

Malicious attacks, including data poisoning, model poisoning, and Byzantine corruption, are considered out of scope.

2) Roles of SecAgg$^+$ and Differential Privacy: Table IV emphasizes that Secure Aggregation and Differential Privacy address orthogonal vulnerabilities and cannot be substituted for one another.

3) Privacy Unit Granularity: By default, ε protects a single network packet record. For FedDP-Marginal, two higher-order privacy units are formalized:

1) **TCP Stream (`-strm{m}`):** Protects entire bidirectional TCP sessions;
2) **Capture Group (`-cap{m}`):** Protects complete contiguous traffic captures.

These guarantees hold strictly under two operational constraints: each unit is hosted entirely on a single client node, and each client clamps contributions to at most m packets per unit. Under these conditions, releasing T contingency tables incurs global sensitivities $\Delta_1 = m \cdot T$ in ℓ_1 and $\Delta_2 = m\sqrt{T}$ in ℓ_2 (`group_dp.py`). In contrast, under packet-level DP, an attack session spanning k packets is protected only via group privacy bounds, which scale weakly to approximately $k\varepsilon$ [49]. The training dataset encompasses 76,463 TCP streams and 147 capture groups (ranging between 62 and 1,277 packets per capture).

4) Public Side Information: The following metadata is considered public and excluded from privacy budget accounting:

- Attack class label names and global sample quotas;
- Client sample counts n_i when transmitted in cleartext (e.g., Flower's `num_examples` in CVAE routes; held static in FedDP-Marginal);
- Preprocessing normalization parameters, logarithmic transformation choices, and categorical encodings computed centrally on pooled data prior to partitioning;
- Hyperparameter selections obtained from the validation split.

D. Experimental Configurations and Robustness Matrix

1) Experiment Matrix: Table V outlines all evaluated configurations across three random seeds $\{0, 1, 2\}$. Configuration naming conventions adopt explicit suffixes:

- `-strm{m}`: Enforces stream-level DP clamping to m packets;
- `-cap{m}`: Enforces capture-level DP clamping to m packets;
- `-t{t}`: Calibrates noise assuming at least t honest clients contribute (default $t = K = 5$; variants evaluate $t \in \{2, 3\}$).

The primary configurations representing PP-FedData across privacy regimes are designated as: `MG-eps1` = `MGr-eps1`, `MG-eps5` = `MGs-eps5`, and `MG-eps10` = `MGs-eps10`, all configured at $\delta = 10^{-5}$.

2) Hyperparameter Optimization: Hyperparameters are tuned exclusively on the held-out validation split using Optuna [52] with the Tree-structured Parzen Estimator (TPE) [53]. Search spaces are segmented by model route (unperturbed CVAE, proxy-scale DP CVAE, full-scale DP CVAE, FedDP-Marginal, and downstream classifiers), as detailed in Section III-K.

3) Robustness and Sensitivity Branches: Table VI details sensitivity excursions designed to test framework stability under non-standard distributions.

E. Requirement-Driven Recommendation Framework

1) Motivation and Relation to Heuristic R6: Standard benchmark studies typically rely on single static selection heuristics.

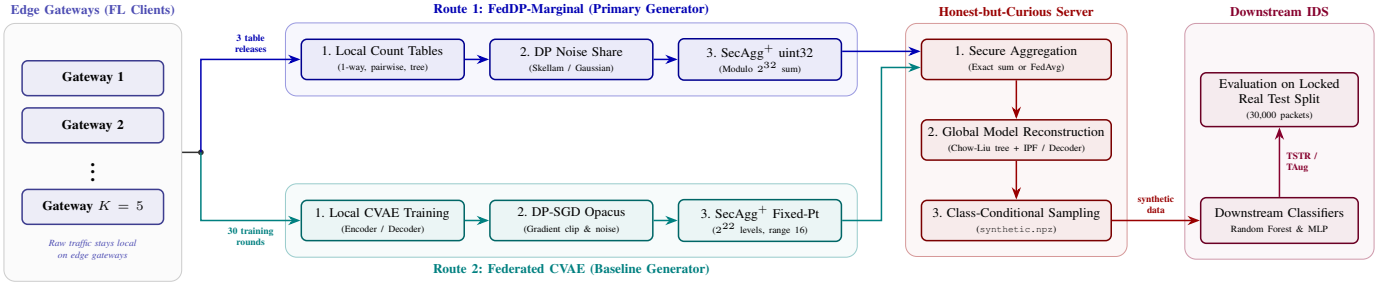


Fig. 2. System architecture of the PP-FedData framework, depicting data flows across edge gateways, dual generator routes, the aggregation server, and downstream evaluation.

TABLE III
SYSTEMATIC COMPARISON OF THE TWO GENERATIVE ROUTES

Architectural Property	FedDP-Marginal (Primary Generator)	Federated CVAE (Baseline Generator)
Shared Artefacts	Contingency count tables summed across clients in 3 releases (1-way marginals, pairwise tables, tree edges).	Model parameter weights exchanged across 30 FedAvg rounds with 2 local epochs per round.
Probabilistic Structure	Chow–Liu dependency tree [10]; optional 2nd-order PrivBayes Bayesian network [9] (MGb).	Deep conditional autoencoder [42] with fully-connected encoder and block-wise decoder.
DP Perturbation Point	Added directly to client tables once per release (discrete Skellam [22] or Gaussian share).	Injected into per-sample clipped gradients at every local step via Opacus [45] (DP-SGD).
Secure Aggregation	Flower SecAgg$^+$ over exact integers (uint32, modular arithmetic $\mathbb{Z}_{2^{32}}$).	Flower SecAgg$^+$ over fixed-point quantised weights (2^{22} levels, dynamic range $[-16, 16]$).
Distributed DP Validity	Formally proven: Each release is a single additive sum of bounded sensitivity.	Not claimed: Multi-epoch non-linear optimization violates linear noise superposition.
Core Implementation	`models/marginal.py`, `fl/mg_app.py`, `group_dp.py`	`models/cvae.py`, `fl/core.py`, `fl/app.py`, `fl/dp_utils.py`

TABLE IV
COMPARISON OF CRYPTOGRAPHIC AND DIFFERENTIAL PROTECTION

Mechanism	Operational Role and Guarantees
Secure Aggregation (SecAgg$^+$)	**Threat Addressed:** Server inspecting individual updates. **Guarantee:** Server learns only the masked sum $\sum \mathbf{x}_i$ [18], [19]. **Limitation:** Zero protection against inferences drawn from the aggregate itself.
Differential Privacy (DP)	**Threat Addressed:** Inference regarding individual units from released models or data. **Guarantee:** (ε, δ)-DP bounds release distribution shift by $e^{\varepsilon} + \delta$ [48]. **Limitation:** Does not conceal raw client updates prior to aggregation unless paired with SecAgg.

For instance, heuristic rule R6 selects the configuration maximizing Random Forest TAug Macro-F1 subject to membership inference AUC ≤ 0.55, worst-case $\varepsilon \leq 5$, and per-round execution time $\leq 3\times$ baseline B3. However, static rules cannot adapt to heterogeneous real-world deployments with varying bandwidth, mutual trust, or threat priorities. PP-FedData introduces a requirement-driven decision engine (`ppfeddata recommend`) operating over an 11-dimensional requirement tuple:

$$\mathcal{R} = \langle \text{Trust}_{\text{srv}}, \text{Trust}_{\text{cli}}, \varepsilon_{\max}, \text{BW}_{\text{rnd}}, \text{BW}_{\text{tot}}, \text{Rounds}_{\max}, \text{Unit}, \text{Priority}, \text{Recall}^{\min}_{\text{rare}}, \text{MIA}^{\max}, h \rangle \quad (1)$$

2) Five-Stage Recommendation Procedure: Operating over empirical scorecard metrics extracted directly from the immutable run ledger, the selection algorithm executes five successive filters:

1) **Validity Filter:** Excludes configurations whose operational properties diverge from verified ledger execution;
2) **Trust Filter:** If the server is untrusted, only configurations supporting SecAgg$^+$ or Local DP are retained. If client collusion is possible, distributed DP is audited at h honest clients;
3) **Constraint Pruning:** Eliminates configurations violating specified ceilings on ε, bandwidth, convergence rounds, or MIA leakage, or failing to satisfy rare-class recall floors;
4) **Pareto Optimization:** Extracts non-dominated configurations across utility, privacy, and resource dimensions;
5) **Tie-Breaking:** Selects the configuration maximizing the prioritized performance objective (Macro-F1, Binary F1, or Rare-Class Recall).

Table VII defines ten representative deployment scenarios used to validate the recommendation framework.

TABLE V
COMPLETE EXPERIMENT MATRIX OF THE PP-FEDDATA FRAMEWORK

Code	Description	Shared / Released Artefact	Protection Mechanism	Evaluation Role
B0	Centralized IDS on pooled real data (TRTR)	None (Direct local training)	None	Theoretical reference ceiling
B1a	B0 with balanced class weights (`class_weight="balanced"`)	None	None	Reference for algorithmic class reweighting
B1b	B0 with SMOTE [24] minority oversampling	None	None	Reference for convex feature interpolation
B2	CVAE trained centrally on pooled data	Pooled raw training data	None	Upper generative bound (pools raw traffic)
B3	CVAE with FedAvg, 5 clients, 30 rounds $\times$ 2 epochs	Model parameter weights	None	Federated baseline ($\varepsilon = \infty$)
M1-eps{1,5,10}	B3 with DP-SGD injected at each client	Noisy parameter gradients	DP (per-client local noise)	Quantifies DP accuracy cost on deep route
M2	B3 with Flower SecAgg$^+$ masking	Masked parameter weights	SecAgg$^+$ cryptographic sum	Quantifies SecAgg overhead on deep route
M3-eps5	Joint DP-SGD and SecAgg$^+$ execution	Noisy, masked weights	Hybrid DP and SecAgg$^+$	Demonstrates multi-layered defense
M1o / M3o-eps{1,5,10}	Hyperparameters re-tuned per ε at full scale	As M1, M3	As M1, M3	Optimised benchmark of deep route (Round O1)
MGs-eps{1,5,10}	FedDP-Marginal with Skellam noise and SecAgg$^+$	3 releases of integer count tables	Distributed DP and SecAgg$^+$	Primary proposed generator of PP-FedData
MGr-eps{1,5,10}	MGs with server IPF [50] and fusion post-processing	Identical to MGs	Identical to MGs	Post-processing (consumes zero additional ε)
MGd-eps{1,5,10}	Gaussian noise distributed across clients (simulated)	Noisy count tables	Distributed DP (zCDP [51])	Accounting alternative to discrete Skellam
MGl-eps{1,5,10}	Gateways add full noise independently (Local DP)	Noisy count tables	Local DP [47]	Zero-trust baseline across edge clients
MGb-eps10	FedDP-Marginal with 2nd-order PrivBayes network [9]	Pairwise contingency tables	Distributed DP (simulated)	Tests high-order dependency structures
FedMLP / FedMLPcw	Directly federated MLP classifier (no generator)	Classifier parameter weights	None (or DP/SecAgg)	Non-generative collaborative IDS reference

TABLE VI
ROBUSTNESS AND SENSITIVITY EXCURSIONS

Branch	Perturbation	Execution Status
Heterogeneity	Dirichlet $\alpha \in \{0.1, 10.0\}$	Evaluated for MGs
Client Scaling	$K \in \{10, 20\}$ clients	Evaluated for MGs, MGl
Fine Labels	11-class DoS/DDoS split	Evaluated for MG and Ref
Split Robustness	Alternative capture splits (A4)	Evaluated (2 seeds)
Stream Clamping	Max 20 pkts/stream (A5)	Evaluated for MGs
A2, A3 Branches	Skewed imbalance ratios	Unrun (needs data)

F. Dataset and Data Collection

1) Dataset Description: The primary data source of this study is the DoS/DDoS-MQTT-IoT dataset published by Alatram *et al.* [2] and obtained from the authors' official Kaggle release [54]. The dataset contains benign IoT traffic together with several denial-of-service (DoS) and distributed denial-of-service (DDoS) attacks that abuse the MQTT protocol [55]. Although the release is distributed in several formats, this study uses a structured CSV representation comprising 59,624,826 rows (approximately 13 GB).

Each row represents a single network packet described by 33 feature columns. The Normal subset carries two additional columns, `Label` and `Capture_ID`; the latter assigns the `group_id` used for data splitting, whereas attack traffic derives its `group_id` from the name of its source file. An initial inventory confirmed that the row counts match the expected per-class totals. Table VIII summarizes the distribution of normal traffic and the five attack families.

2) Data Provenance and Verification: Attack traffic is used exactly as published: the 48 attack CSV files exported by the dataset authors [2], whose paths and sizes match the Kaggle release [54]. Because the release provides Normal traffic only as 49 PCAP files, every Normal packet was exported with TShark 4.4.7 [56] into the same 33 columns, using a script maintained in the project repository.

The export is complete, since the number of rows equals the total packet count of all 49 captures (45,580,102 rows), and reproducible, since re-extracting three captures yields byte-identical files. To validate the procedure, it was applied to the attack PCAP files, for which the authors' CSV files serve as ground truth: 23 of the 25 raw features used by the models match the authors' values in at least 99.98% of rows. The two remaining discrepancies are explained by the authors' export settings, namely a display filter that retains only packets sent to the broker and two-pass dissection; enabling both settings reproduces the authors' rows in all 48 captures. The Normal export therefore contains every packet that the authors' settings would select from the same PCAP files, plus the packets sent by the broker; the implications of this difference are discussed in Section III-N. Table IX summarizes the verification checks.

3) Label Harmonization and Rare Classes: Target labels are harmonized into a six-class configuration, namely `NORMAL`, `BCF`, `DELAYED`, `SYN`, `INVALID`, and `WILL`, in which the DoS and DDoS variants of the same attack family are merged

TABLE VII
ILLUSTRATIVE EDGE DEPLOYMENT SCENARIOS AND REQUIREMENT FORMULATIONS

Deployment Scenario	Operational Context and Assumptions	Formal Requirement Vector $\mathcal{R}$
`lab`	Single operator controls aggregation server and all gateways	Server/clients trusted; $\varepsilon =$ None; priority: `macro`
`cloud-aggregator`	Enterprise gateways aggregated on public third-party cloud	Server untrusted; clients trusted; $\varepsilon \leq 5$; priority: `macro`
`consortium`	Competing IoT operators susceptible to mutual collusion	Server untrusted; clients collude; $\varepsilon \leq 10$; priority: `macro`
`strict-privacy`	Regulated environments (e.g., healthcare IoT infrastructure)	Server untrusted; clients trusted; $\varepsilon \leq 1$; priority: `macro`
`low-bandwidth`	Cellular backhaul, LoRaWAN, or NB-IoT connections	Server untrusted; $\varepsilon \leq 10$; $\mathrm{BW}_{\mathrm{rnd}} \leq 2\,\mathrm{MB}$; $\mathrm{BW}_{\mathrm{tot}} \leq 100\,\mathrm{MB}$; priority: `macro`
`rare-attacks`	Critical infrastructure prioritizing rare DoS/DDoS detection	Server untrusted; clients trusted; $\varepsilon \leq 5$; priority: `rare`
`honest-majority`	Multi-operator network with at most two colluding clients	Server untrusted; $h \geq 3$ honest clients; $\varepsilon \leq 10$; MIA AUC < 0.55; priority: `macro`
`gateway-dropout`	Highly volatile network where up to 2 clients drop out	Server untrusted; $h \geq 3$ honest clients; $\varepsilon \leq 5$; priority: `macro`
`session-privacy`	Protection must formally cover entire bidirectional TCP sessions	Server untrusted; $\varepsilon \leq 5$; Unit = stream; priority: `macro`
`capture-privacy`	Protection must cover entire network packet captures	Server untrusted; $\varepsilon \leq 10$; Unit = capture; priority: `macro`

TABLE VIII
COMPOSITION OF THE DOS/DDOS-MQTT-IOT DATASET

Class	DoS	DDoS	Total Rows	Share
NORMAL	—	—	45,580,102	76.44%
BCF	5,825,026	6,455,848	12,280,874	20.60%
DELAYED	291,988	309,554	601,542	1.01%
SYN	103,828	497,611	601,439	1.01%
INVALID	109,150	181,190	290,340	0.49%
WILL	114,474	156,055	270,529	0.45%
Total			**59,624,826**	**100.00%**

BCF: Basic Connect Flooding; DELAYED: Delayed Connect Flooding; SYN: SYN TCP Flooding; INVALID: Invalid Subscription Flooding; WILL: Connect Flooding with WILL payload.

into a single class. This merging is supported by leakage check C5 (Section III-F7), which shows that DoS and DDoS variants within the same family are practically indistinguishable at the packet level. The original 11-class labeling remains available in the data pipeline but is used only for the robustness analysis (Table VI).

Rare classes are defined as attack classes whose training quota does not exceed 5,000 rows, i.e., `DELAYED`, `SYN`, `INVALID`, and `WILL`. Their rarity during training is a deliberate consequence of the quota design rather than a lack of raw data: as shown in Table VIII, the raw dataset contains between 270,529 and 601,542 packets for each of these classes. The quotas impose a 60:1 imbalance between the majority class (`NORMAL`, 60,000 rows) and the smallest minority class (`WILL`, 1,000 rows), which creates a demanding setting for evaluating synthetic data generators under federated constraints [3]. In contrast, the validation and test sets are class-balanced (2,000 and 5,000 rows per class, respectively), so that the evaluation metrics are not dominated by the majority class.

4) Schema Harmonization and Feature Selection: Preprocessing begins by harmonizing the heterogeneous raw packet exports. Column names are converted to snake case. Wireshark text labels (e.g., `message_type`, `qos_level_*`, and `requested_qos`) are normalized to canonical codes, and boolean indicators (True/False, Set/Not set) are mapped to 0/1. Redundant columns with the suffix `_1` are removed. For multi-valued cells that contain several MQTT messages, only the first element is retained, consistent with the Normal export produced with the TShark `occurrence=f` option; the `n_mqtt_msgs` column is therefore discarded. Rows are then filtered to those whose protocol is TCP or MQTT, followed by a format audit of time-based tokens.

To prevent data leakage [57], [58], identifier and metadata columns are removed: `no` (row sequence), `epoch_time` (absolute timestamp), and `time_since_reference_or_first_frame` (which reveals the capture start time). The `stream_index` column is retained only as grouping metadata and is not used as an input feature. The `source` column is removed because sender IP addresses would allow a model to memorize host identities instead of learning traffic patterns. Free-text payloads in the `info` column and the Normal-only `label` and `capture_id` columns are also dropped; the raw dataset contains no MAC address or port columns. After feature selection, the final schema comprises $D = 60$ input features, as detailed in Table X.

5) Group-Based Splitting and Quota Sampling: The 11 fine-grained subclasses (`NORMAL` plus the DoS and DDoS variants of the five attack families) are partitioned with a group-based strategy: entire capture groups are assigned to the training, validation, or test split according to a fixed `split_seed`. This prevents highly correlated packets from the same TCP stream or capture session from appearing in more than one split [58].

TABLE IX
DATA PROVENANCE VERIFICATION

Validation Question	Verification Method	Result
Are the files authentic to the release?	Compared file paths and sizes against the public Kaggle manifest [54].	411/411 files match (49 attack CSVs, 313 attack PCAPs, 49 Normal PCAPs).
Does the Normal CSV contain all packets?	Compared row counts with PCAP packet counts reported by `capinfos` [56]; verified sequential frame numbers $1, \ldots, N$.	Complete match across 49/49 captures (45,580,102 rows).
Is the extraction reproducible?	Re-extracted three captures and compared SHA-256 digests.	Files are byte-identical.
Does the procedure yield correct feature values?	Applied the export script to 48 attack PCAPs and compared the output row by row with the authors' CSVs.	23/25 model features match in $\geq 99.98\%$ of rows.
What explains the remaining discrepancies?	Traced to the authors' two-pass dissection and display filter.	Enabling both settings reproduces the authors' rows; the residual 0.04% differences stem from Wireshark version changes.

TABLE X
FINAL FEATURE SCHEMA ($D = 60$)

Feature Type	Columns	Dim.
Numeric	`time_delta_from_previous_displayed_frame`, `frame_length_on_wire_2`, `irtt`, `time_since_first_frame_in_this_tcp_stream`, `tcp_segment_len`, `calculated_window_size`, `keep_alive`, `user_name_length`, `password_length`, `will_message_length`, `will_topic_length`, `topic_length`, `msg_len`	13
Binary	`syn`, `reset`, `acknowledgment`, `retain`, `clean_session_flag_2`, `will_retain`, `will_flag`	7
Missingness flags	12 feature-specific `_is_na` indicators	12
Categorical (one-hot)	`message_type` (12), `qos_level_1` (5), `qos_level_2` (4), `requested_qos` (3), `protocol` (4)	28
Total		**60**

However, 8 of the 11 subclasses originate from a single capture file, which precludes a purely group-based assignment. For these subclasses, the validation and test sets are extracted as contiguous blocks of the same capture used for training (*block split*), separated by 1,000-row buffers to mitigate adjacency leakage. TCP streams that cross a block boundary are discarded entirely to preserve stream integrity. This removal eliminates between 1.2% and 9.9% of the eligible rows within each affected subclass and introduces a slight sampling bias toward shorter connections.

After splitting, rows are drawn by random sampling without replacement according to the class quotas in Table XI, which emulate the data scarcity faced by individual gateways. The training quotas enforce severe imbalance, whereas the validation and test quotas are balanced. The 30,000-row test set is locked and used only for final reporting; all hyperparameter tuning, design choices, and threshold calibration rely exclusively on the validation set.

6) Preprocessing and Encoding: The preprocessing pipeline is fitted on the training split only; the validation and test sets are transformed with the stored parameters to prevent leakage. Missing or non-applicable numeric values are imputed with 0, and a binary `_is_na` indicator is added for every feature whose missing rate in the training set lies strictly between 0.005 and 0.995.

TABLE XI
CLASS QUOTAS FOR THE TRAINING, VALIDATION, AND TEST SPLITS

Class	Train	Validation	Test
NORMAL	60,000	2,000	5,000
BCF	20,000	2,000	5,000
DELAYED	3,000	2,000	5,000
SYN	3,000	2,000	5,000
INVALID	1,500	2,000	5,000
WILL	1,000	2,000	5,000
Total	**88,500**	**12,000**	**30,000**

Numeric features with a skewness greater than 2 are transformed with $\log(1 + x)$; time-delta features are first multiplied by 1,000 to convert seconds into milliseconds. Numeric columns are then standardized to zero mean and unit variance and clipped at $\pm 5\sigma$ to limit the influence of extreme outliers. Categorical features are one-hot encoded, retaining the ten most frequent training tokens per column; the remaining valid values are mapped to an `OTHER` token and missing values to a `NONE` token.

Normalization statistics, categorical vocabularies, and imputation parameters are fitted centrally on the pooled training set. This constitutes a methodological limitation: it assumes a centralized view of the data prior to federation, and these statistics

TABLE XII
LEAKAGE AND LABEL-RELIABILITY CHECKS

Code	Diagnostic Check	Flag Threshold
C1	Cross-class column presence analysis to identify attributes structurally bound to specific attacks.	Column empty in one class but populated in another.
C2	Random Forest trained only on the `_is_na` missingness flags.	Macro-F1 > 0.5.
C3	Class-balanced Random Forest trained on individual raw columns, plus ablations omitting specific features.	Macro-F1 or best per-class F1 > 0.9 on any single feature.
C4	Generalization gap between a random row split and the group-based stream split.	Difference (random $-$ group) > 0.05.
C5	Binary classification of DoS versus DDoS variants within the same attack family.	F1 < 0.6.

TABLE XIII
FEDERATED PARTITIONING DESIGN PARAMETERS

Parameter	Specification	Role / Scope
Gateways K	5 (canonical); $\{10, 20\}$	Federation size; scalability study
Concentration α_{dir}	0.5 (canonical); $\{0.1, 10\}$	Non-IID benchmark; robustness sweep
Mechanism	Per-class Dirichlet [40]	Row-based (record DP); unit-based (group DP)
Resampling floor	$n_k \geq 500$ records	Rejection threshold, record level
Unit floor	$n_k \geq 200$ records	Rejection threshold, unit level
Seeds	$\{0, 1, 2\}$	Independent draws shared across models
Training quota	$N = 88{,}500$ (Table XI)	Global class distribution partitioned across gateways

lie outside the differential privacy budget ε (Section III-C).

7) Leakage and Label-Reliability Checks: Before synthetic data generation, a suite of diagnostic checks verifies that the learning task is not artificially simplified by schema artefacts and that the attack labels are reliable. All checks use a Random Forest [41] with 100 trees, evaluated on the validation set over three seeds $\{0, 1, 2\}$. The suite comprises five checks (C1–C5), each associated with a heuristic threshold that flags anomalies requiring manual inspection or schema adjustment, as listed in Table XII.

G. Federated Partitioning

1) Dirichlet Label-Skew Partitioning: To model non-identically distributed (non-IID) traffic across edge gateways while preserving protocol coherence, the centralized training set is partitioned across a federation of gateways $\mathcal{K} = \{1, 2, \ldots, K\}$, with $K = 5$ in the canonical benchmark. Partitioning follows the label-distribution skew model of Hsu *et al.* [40], parameterized by a symmetric Dirichlet distribution with concentration parameter α_{dir}. For each class $c \in \{1, \ldots, C\}$, with $C = 6$ in the canonical label configuration, the assignment proportions across the K gateways are drawn independently as

$$\mathbf{p}_c = (p_{c,1}, \ldots, p_{c,K}) \sim \text{Dir}(\alpha_{\text{dir}} \mathbf{1}_K), \tag{2}$$

where $\mathbf{1}_K = (1, \ldots, 1) \in \mathbb{R}^K$. The concentration parameter controls statistical heterogeneity [40]: small values ($\alpha_{\text{dir}} \to 0$) concentrate each class on few gateways, whereas large values ($\alpha_{\text{dir}} \to \infty$) converge toward an IID allocation. The canonical benchmark adopts $\alpha_{\text{dir}} = 0.5$, under which some gateways naturally receive very few or no instances of the rare attack classes.

2) Record-Level versus Unit-Level Partitioning: Two partitioning regimes are distinguished:

- **Record-Level Partitioning (Default):** Each packet record (x, y) of class c is assigned to gateway k with probability $p_{c,k}$. Packets from the same TCP stream or capture group may therefore span several gateways, which models an open edge network in which packets of distributed flows traverse different regional edge routers.
- **Unit-Level Partitioning (Group DP):** Reserved for session-level and capture-level differential privacy evaluations (Section III-I4). TCP streams or capture groups are treated as indivisible units: Dirichlet proportions are applied according to the majority class of each unit, and all records of a unit are assigned to a single gateway. This guarantees that no protected entity is fragmented across gateways, giving an unambiguous boundary for group-level contribution bounding.

3) Partition Constraints and Safeguards: The partitioning protocol enforces the following constraints to avoid degenerate simulation states:

- **Resampling Safeguard:** If any gateway receives fewer than 500 records in total ($\sum_c n_{k,c} < 500$), the sampled proportions are rejected and redrawn from $\text{Dir}(\alpha_{\text{dir}} \mathbf{1}_K)$ until all gateways satisfy the floor;
- **Unit-Level Floor:** Under unit-level partitioning, a minimum of 200 records per gateway is enforced so that every node holds sufficient local samples for valid aggregation;
- **Seed-Specific Realization:** Each seed $s \in \{0, 1, 2\}$ instantiates an independent partition draw $\mathbf{p}_c^{(s)}$, and all architectures evaluated on seed s share the same partition instance, which removes partitioning as a confounder;
- **Pre-Partition Split Isolation:** The global train/validation/test split is executed before federated allocation, so the validation and test sets never enter the decentralized training pool. Full session independence across splits remains limited by the block split of single-capture subclasses (Section III-F and Section III-N).

The partition design parameters are summarized in Table XIII.

4) Partition Invariance of Linear Marginal Aggregation: For any contingency query aggregated under SecAgg$^+$ [18], the global result is an exact unweighted sum of local integer count vectors:

$$\mathbf{h}_{\text{global}} = \sum_{k=1}^{K} \mathbf{h}_k = \sum_{k=1}^{K} \sum_{i \in \mathcal{D}_k} \mathbf{e}(x_i) = \sum_{i \in \bigcup_k \mathcal{D}_k} \mathbf{e}(x_i), \tag{3}$$

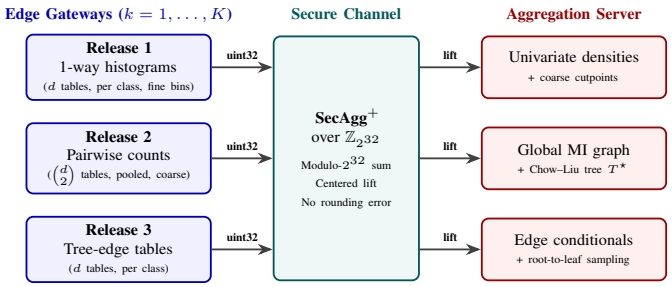


Fig. 3. Three-release protocol of FedDP-Marginal. Coarse cutpoints (after Release 1) and the tree $T^\star$ (after Release 2) are broadcast back to the gateways before the next release.

where $\mathbf{e}(x_i)$ is the standard basis indicator vector of the bin containing record x_i. Because integer addition is commutative and associative, the unperturbed aggregated table is algebraically identical to the centralized table, irrespective of label skew $P_k(Y)$ or covariate shift $P_k(X \mid Y)$.

Evaluating FedDP-Marginal across $\alpha_{\text{dir}} \in \{0.1, 0.5, 10\}$ therefore serves as a *null prediction* (sanity check): without client dropout or entity capping, its aggregated marginals and synthesis fidelity are theoretically invariant to partition skew. In operational networks, heterogeneity affects FedDP-Marginal only through two edge conditions:

1) **Client dropout:** The counts of dropped gateways are omitted from the sum, so local class samples that are absent from surviving gateways under small α_{dir} are lost;
2) **Record-level splits under caps:** With a group contribution cap m, an entity spanning several gateways is clipped independently up to m times per gateway, whereas on a single gateway it would be clipped only once.

H. FedDP-Marginal Generator

1) Overview: FedDP-Marginal adapts private marginal-based tabular synthesis, specifically the MST framework of McKenna *et al.* [59] and the graphical-model principles of PrivBayes [9], to a federated architecture. It is not a new generative foundation; rather, it adapts marginal-based dependency modeling to the federated setting through secure integer aggregation. Unlike deep federated models that require dozens of rounds over model weights, FedDP-Marginal uses an exact three-release query protocol coordinated via SecAgg$^+$ [18] in Flower [43], as illustrated in Fig. 3.

2) Attribute Schema: The marginal synthesizer and the neural baseline use different feature representations:

- **Tabular schema ($d = 37$ attributes):** FedDP-Marginal queries contingency tables over 13 numeric attributes, discretized into $B_{\text{fine}} = 64$ uniform bins over the normalized interval $[-5, 5]$ (consistent with the $\pm 5\sigma$ clipping); 19 binary attributes, comprising 7 protocol flags (`syn`, `reset`, `acknowledgment`, `retain`, `clean_session_flag_2`, `will_retain`, `will_flag`) and 12 `_is_na` indicators; and 5 categorical attributes (`message_type`, `qos_level_1`, `qos_level_2`, `requested_qos`, `protocol`) represented directly as discrete variables;
- **One-hot representation ($D = 60$):** The CVAE baseline operates on $\mathbb{R}^D$, obtained by expanding the 5 categorical attributes into 28 one-hot indicators ($13 + 19 + 28 = 60$; Table X).

3) Three-Release Protocol: The protocol issues three sequential releases that allocate the privacy budget across complementary granularities:

- **Release 1 (fine univariate marginals per class):** Each gateway tabulates 1-way counts for every attribute X_i, $i \in \{1, \ldots, d\}$, conditioned on the class label c over $B_{\text{fine}} = 64$ bins, adds its discrete noise share, and submits the result through SecAgg$^+$ [18]. From the aggregated noisy histograms, the server sets data-dependent coarse cutpoints for numeric attributes at empirical quantiles, with $B_{\text{coarse}} \in \{4, 8, 16, 32\}$ (default 16); coarse cutpoints are aligned with fine bin edges.
- **Release 2 (coarse pairwise marginals pooled across classes):** Gateways compute 2-way tables for all $\binom{d}{2} = 666$ attribute pairs over coarse bins, pooled across all classes. Pooling maximizes the count per cell and bounds table size to B_{coarse}^2, which limits the relative impact of privacy noise on dependency estimation.
- **Release 3 (tree-edge marginals conditioned on class):** Using the structure derived from Release 2, the server requests joint tables for the $d - 1$ edges of the global Chow–Liu tree conditioned on $y = c$ over coarse bins (cardinality $B_{\text{coarse}}^2 \cdot C$, which avoids noise swamping on rare classes), together with the class-conditional root table over fine bins.

4) Single Global Chow–Liu Tree: FedDP-Marginal constructs a *single* global Chow–Liu dependency tree [10] pooled across classes rather than one tree per class. From the post-processed Release 2 tables, the server computes the empirical mutual information (MI) between attributes and extracts the maximum spanning tree with Kruskal's algorithm [60]:

$$T^\star = \arg\max_{T} \sum_{(i,j)\in T} \hat{I}(X_i; X_j), \tag{4}$$

$$\hat{I}(X_i; X_j) = \sum_{a,b} \hat{p}(a,b) \log \frac{\hat{p}(a,b)}{\hat{p}(a)\,\hat{p}(b)}. \tag{5}$$

Following the post-processing approach of McKenna *et al.* [59], empirical probabilities are obtained from the noisy pairwise counts $c_{i,j}$ by non-negative thresholding, additive (Laplace) smoothing α_{sm}, and normalization:

$$\hat{p}(a,b) = \frac{\max(0, c_{i,j}(a,b)) + \alpha_{\text{sm}}}{\sum_{a',b'} [\max(0, c_{i,j}(a',b')) + \alpha_{\text{sm}}]}, \tag{6}$$

and the univariate terms $\hat{p}(a)$ and $\hat{p}(b)$ are obtained by summing out the corresponding dimension. By the post-processing property of differential privacy [49], deriving $T^\star$ consumes no additional privacy budget.

Pooling trade-off. Class-conditional MI $\hat{I}(X_i; X_j \mid Y = c)$ would align more closely with class-conditional synthesis.

However, splitting Release 2 by class leaves the per-cell sensitivity unchanged while dividing the per-cell signal by roughly C, and the total noise mass across the $C\binom{d}{2}$ tables grows C-fold; the per-cell signal-to-noise ratio therefore degrades by a factor of C and corrupts edge selection. FedDP-Marginal deliberately pools pairwise counts to maximize the signal-to-noise ratio of the skeleton $T^\star$. The pooled MI thus reflects the overall dependency structure, which is dominated by `NORMAL` traffic (67.8% of the training quota), while class-specific relationships are captured by the class-conditional edge tables of Release 3.

Cardinality bias. Plug-in MI on noisy tables has a finite-sample bias proportional to the table degrees of freedom $(|X_i|-1)(|X_j|-1)$: up to $(16-1)^2 = 225$ for pairs of numeric attributes with $B_{\text{coarse}} = 16$, compared with 1 for binary pairs. Kruskal's algorithm on unnormalized MI therefore tends to favor edges between high-cardinality numeric attributes, and additive smoothing, which adds pseudo-mass $\alpha_{\text{sm}}|X_i||X_j|$, dilutes probabilities more strongly in larger tables. The Miller–Madow correction [61]

$$\hat{I}_{\text{MM}}(X_i; X_j) = \hat{I}(X_i; X_j) - \frac{(|X_i|-1)(|X_j|-1)}{2\hat{N}}, \quad (7)$$

with $\hat{N} = \sum_{a,b} \max(0, c_{i,j}(a,b))$, adjusts for asymptotic multinomial sampling bias but does not account for additive Skellam noise or the non-linear truncation $\max(0,\cdot)$. The canonical pipeline retains uncorrected plug-in MI with $B_{\text{coarse}} \le 16$ as the baseline heuristic. To quantify the effect of cardinality bias on structure discovery separately from downstream utility, an ablation comparing plug-in MI, Miller–Madow MI, and normalized MI ($\hat{I}(X_i;X_j)/\sqrt{H(X_i)H(X_j)}$), together with a tree-edge stability check across partition seeds, is included in search space (d) (Section III-K).

5) Class-Conditional Sampling: Synthetic records of class c are generated by traversing $T^\star$ in topological root-to-leaf order [10]:

$$\hat{P}(\mathbf{x} \mid y=c) = \hat{P}(x_r \mid y=c) \prod_{(u,v)\in T^\star} \hat{P}(x_v \mid x_u, y=c), \quad (8)$$

where r denotes the root attribute. Sampling proceeds as follows:

1) **Root selection and sampling:** For MGs, the root is the attribute maximizing $\sum_{j\neq r} \hat{I}(X_r; X_j)$; it is sampled from its class-conditional fine distribution of Release 1, $x_r \sim \hat{P}(X_r \mid y=c)$;
2) **Conditional propagation:** Each child is sampled given its parent, $x_v \sim \hat{P}(X_v \mid X_u = x_u, y=c)$, from noisy counts that are thresholded at zero and smoothed with α_{sm};
3) **Continuous refinement:** For numeric attributes, sampled coarse bins are refined to fine bins in proportion to the fine class histogram of Release 1 (whose bins nest within the coarse bins), followed by uniform interpolation within $[b_{\text{low}}, b_{\text{high}}]$;
4) **Pipeline inversion:** The resulting vectors pass through the same inverse transformation (inverse standardization and inverse $\log(1+x)$) as the CVAE baseline.

6) Sensitivity, Budget Allocation, and Consistency Refinement: Neighboring relation and sensitivity. Accounting uses unbounded adjacency, i.e., datasets $D \sim D'$ differing by the addition or removal of one packet record. A single record increments exactly one cell in each of the d tables of Release 1, the $\binom{d}{2}$ tables of Release 2, and the d tables of Release 3 ($d-1$ edge tables plus the root table). For a release of T tables, the global sensitivities are therefore $\Delta_1 = T$ and $\Delta_2 = \sqrt{T}$. Under bounded DP (record replacement within a class), they become $\Delta_1^{\text{repl}} = 2T$ and $\Delta_2^{\text{repl}} = \sqrt{2T}$.

Budget split. The budget is distributed across releases by a split vector $\mathbf{s} = (s_1, s_2, s_3)$ with $s_1 + s_2 + s_3 = 1$, defaulting to $(0.3, 0.1, 0.6)$. For the Gaussian variant (MGd), the zCDP parameter is split as $\rho_r = s_r\rho$, giving per-release noise standard deviation $\sigma_r = \Delta_{2,r}/\sqrt{2s_r\rho}$. For the Skellam variant (MGs), the noise parameter of release r is $\mu_{\text{sk},r} = \Delta_{2,r}^2/(2s_r\kappa)$, where the global factor $\kappa > 0$ is found by binary search. Under the default split, Release 2 has $\Delta_{2,2}^2 = \binom{37}{2} = 666$ with $s_2 = 0.1$ and Release 1 has $\Delta_{2,1}^2 = 37$ with $s_1 = 0.3$, so $\mu_{\text{sk},2}/\mu_{\text{sk},1} = (666/0.1)/(37/0.3) = 54$. Release 2 thus carries about $54\times$ the per-cell noise variance of Release 1, which makes coarse binning ($B_{\text{coarse}} \le 16$) necessary to preserve the signal-to-noise ratio.

Consistency refinement (MGr). To enforce consistency without consuming privacy budget, MGr applies the following post-processing steps:

1) **Inverse-variance count merging:** Merging is performed in count space per class before normalization. Fine Release 1 counts are collapsed to coarse bins; a coarse bin b of X_u containing k_b fine bins has variance $k_b\sigma_1^2$. Projecting an incident Release 3 edge table (X_u, X_v) onto X_u by summing over the $|X_v|$ coarse bins of X_v yields an independent estimate with variance $|X_v|\sigma_3^2$. A non-root attribute with degree $d(u)$ in $T^\star$ thus has $1+d(u)$ independent estimators; the root additionally has the class-conditional fine root table of Release 3 (variance $k_b\sigma_3^2$) and merges $2+d(r)$ estimators. Following McKenna *et al.* [59], estimators are merged bin-wise as

$$\hat{C}_u^{\text{mrg}}(b \mid c) = \frac{\sum_j w_{u,j}(b)\, \hat{C}_{u,j}(b \mid c)}{\sum_j w_{u,j}(b)}, \quad (9)$$

with $w_{u,j}(b) = 1/\text{Var}(\hat{C}_{u,j}(b \mid c))$, and then normalized to probabilities $\hat{P}_u^{\text{mrg}}(b) = \hat{C}_u^{\text{mrg}}(b)/\sum_{b'} \hat{C}_u^{\text{mrg}}(b')$;
2) **Non-negative truncation:** Merged marginals and noisy Release 3 tables are projected onto the non-negative orthant and normalized;
3) **Iterative proportional fitting (IPF):** Each edge table $(X_u, X_v \mid y=c)$ is adjusted with IPF [50], [62] to match the merged marginals of X_u and X_v, which guarantees margin consistency;

4) **Smoothing and root selection:** Additive smoothing $\alpha_{\text{sm}} \in \{0.1, 1, 10\}$ is applied, and the root is selected by the maximum sum of noisy MI.

Second-order Bayesian network (MGb). This ablation adapts PrivBayes [9]: parent pairs (u_1, u_2) are selected greedily from Release 2 MI subject to acyclicity. The number of tables touched per record remains d, so record sensitivity is unchanged, while individual tables grow to accommodate two parents.

7) Secure Transport via SecAgg$^+$: For MGs and MGr, updates are transmitted through Flower SecAgg$^+$ [18] over the integer ring $\mathbb{Z}_M$ with $M = 2^{32}$ (`uint32`):

- Client updates use `max_weight` with unit scaling, and the quantization range is set to twice the clipping range, which guarantees exact integer transmission without floating-point rounding error;
- Under record-level DP, `clipping_range` $= 100{,}000$, which exceeds the largest possible cell count ($N = 88{,}500$). Under group DP, it is set to $C_{\text{clip}} = N + 6\sqrt{\mu_{\text{sk}}^{\text{group}}/K}$, which bounds both the data mass and the 6σ tail of the discrete noise, so that the probability of a perturbed count exceeding C_{clip} is below 10^{-9};
- Perturbed counts $c_{k,j} = x_{k,j} + \xi_{k,j} \pmod{2^{32}}$ are recovered at the server by centered lifting, $\text{lift}(S) = S$ if $S < 2^{31}$ and $S - 2^{32}$ otherwise. Since $\sum_k x_{k,j} \le N$ and, under record-level DP, $\sigma_{\text{agg}} = \sqrt{\mu_{\text{sk}}} \le 400$, overflow is impossible in practice. Under group DP with cap $m = 100$, $\mu_{\text{sk}}^{\text{group}} \le 1.6 \times 10^9$, so $\sigma_{\text{agg}} \le 4 \times 10^4$ (per-client share $\sigma_k = \sqrt{\mu_{\text{sk}}/K} \le 1.8 \times 10^4$), and $|S| \le N + 6\sigma_{\text{agg}} \approx 3.3 \times 10^5 \ll 2^{31} \approx 2.15 \times 10^9$; the probability of modular wrap-around is below 10^{-50};
- The Gaussian variant MGd is simulated in-process over $\mathbb{R}$, and the Local DP variant MGl transmits unshared noisy tables without SecAgg$^+$.

Table XIV summarizes the FedDP-Marginal variants.

I. Privacy Accounting

1) Analytical zCDP Accounting for the Gaussian Comparator (MGd): MGd is accounted with analytical zero-concentrated differential privacy (zCDP) [51]. For release $r \in \{1, 2, 3\}$ with noise standard deviation σ_r and ℓ_2 sensitivity $\Delta_{2,r}$, the accumulated parameter ρ and the resulting (ε, δ) guarantee are [51]

$$\rho = \sum_{r=1}^{3} \frac{\Delta_{2,r}^2}{2\sigma_r^2}, \qquad \varepsilon = \rho + 2\sqrt{\rho \ln(1/\delta)}, \tag{10}$$

with ρ split across releases by $\mathbf{s} = (0.3, 0.1, 0.6)$ by default. The analytical zCDP conversion is intrinsically looser than numerical RDP conversion [63]; at identical nominal ε, utility differences between MGd and MGs therefore partly reflect the tightness of the accountant rather than the noise mechanism alone.

2) Skellam Mechanism and Accounting (MGs, MGr): Nominal guarantees are evaluated at $\varepsilon \in \{1, 5, 10\}$ with $\delta = 10^{-5}$. The Skellam distribution $\text{Sk}(\mu_{\text{sk}}) = \text{Pois}(\mu_{\text{sk}}/2) - \text{Pois}(\mu_{\text{sk}}/2)$ is the difference of two independent Poisson variables, with zero mean and variance μ_{sk} [22], [64]. Because it is closed under summation, integer noise shares generated by the gateways add up exactly under modular addition over $\mathbb{Z}_{2^{32}}$ [22]. Accounting uses the Rényi differential privacy (RDP) bound of Agarwal *et al.* [22, Corollary 3.6]: for integer order $\lambda > 1$,

$$\varepsilon(\lambda) \le \frac{\lambda \Delta_2^2}{2\mu_{\text{sk}}} + \min\left(\frac{(2\lambda - 1)\Delta_2^2 + 6\Delta_1}{4\mu_{\text{sk}}^2}, \frac{3\Delta_1}{2\mu_{\text{sk}}} \right). \tag{11}$$

RDP is composed additively over the three releases [65], $\varepsilon_{\text{total}}(\lambda) = \sum_{r=1}^{3} \varepsilon_r(\lambda)$, and converted to (ε, δ) with the numerical RDP accountant of Opacus [45]. Because (11) holds only for integer orders, the accountant grid is restricted to $\lambda \in \{2, 3, \ldots, 128\}$. A global calibration factor for μ_{sk} is found by binary search so that the composed guarantee matches the target (ε, δ).

3) Trust Models and Noise Partitioning: Noise calibration is evaluated under three trust assumptions:

- **Distributed DP ($t = K$, canonical):** Each gateway injects $1/K$ of the total variance, $\mu_{\text{sk},k} = \mu_{\text{sk}}/K$. The nominal guarantee holds when all K gateways are honest; when only $h < K$ remain honest, the effective $\varepsilon(h)$ is recomputed with the Opacus RDP accountant from the reduced aggregate variance $\frac{h}{K}\mu_{\text{sk}}$, $h \in \{1, \ldots, K\}$.
- **Threshold DP ($t \le K$):** Each gateway injects $\mu_{\text{sk},k} = \mu_{\text{sk}}/t$, so the target guarantee holds whenever at least t gateways participate honestly. Under SecAgg$^+$, masks are shared via Shamir secret sharing with reconstruction threshold T_{recon} [18]; to prevent colluding gateways from reconstructing honest gateways' masking seeds, the number of colluders $c = K - t$ must satisfy $c < T_{\text{recon}}$. The evaluated configurations are: $t = 5$ (canonical distributed DP, $c = 0$, $T_{\text{recon}} = 3$); $t = 3$ (canonical threshold DP, $c \le 2 < T_{\text{recon}} = 3$, tolerating up to two dropouts or two semi-honest colluders while preserving aggregate-only visibility); and $t = 2$ (hardened threshold DP), which tolerates $c = 3$ colluders only if T_{recon} is raised to 4 (leaving one dropout of tolerance), since with $T_{\text{recon}} = 3$ three colluders could reconstruct honest masks and reduce the guarantee to Local DP.
- **Local DP (MGl):** Each gateway injects the full variance, $\mu_{\text{sk},k} = \mu_{\text{sk}}$. No trust in peers or the server is required and SecAgg$^+$ is not used, but the aggregate carries $K\mu_{\text{sk}}$ noise variance.

4) Group Privacy and Contribution Bounding: To protect against session tracing or capture re-identification, group differential privacy is enforced by scaling sensitivities with the per-unit record cap m [49]:

$$\Delta_1^{\text{group}} = m\Delta_1, \qquad \Delta_2^{\text{group}} = m\Delta_2. \tag{12}$$

The Gaussian standard deviation then scales as $\sigma^{\text{group}} = m\sigma$ and the Skellam parameter as $\mu_{\text{sk}}^{\text{group}} \approx m^2\mu_{\text{sk}}$. Group DP holds formally when (i) each protected unit (TCP stream or capture group) resides entirely on one gateway through unit-level partitioning, and (ii) each gateway caps every unit to at

TABLE XIV
ARCHITECTURAL VARIANTS OF FEDDP-MARGINAL

ID	Noise	Privacy Target	Aggregation Channel	Experimental Function
MGs	Skellam	Distributed (ε, δ)	SecAgg$^+$ over $\mathbb{Z}_{2^{32}}$	Primary proposed generator (additive noise sharing under SecAgg$^+$)
MGr	Skellam	Distributed (ε, δ)	SecAgg$^+$ over $\mathbb{Z}_{2^{32}}$	MGs with inverse-variance merging and IPF consistency refinement
MGd	Gaussian	Distributed (ε, δ)	In-process simulation over $\mathbb{R}$	Continuous Gaussian comparator (analytical zCDP)
MGl	Skellam / Gaussian	Local DP (ε, δ)	Direct sum (no SecAgg$^+$)	Local DP baseline (sum of K unshared perturbations)
MGb	Skellam	Distributed (ε, δ)	SecAgg$^+$ over $\mathbb{Z}_{2^{32}}$	Second-order Bayesian network ablation (PrivBayes style)
MGs-nodp	None ($\varepsilon = \infty$)	None	SecAgg$^+$ over $\mathbb{Z}_{2^{32}}$	Non-private reference under SecAgg$^+$ (counterpart of M2)
MGp-nodp	None ($\varepsilon = \infty$)	None	Plain Flower FedAvg	Non-private plaintext reference (counterpart of B3)

most m records. Under default record-level DP ($m = 1$), a unit of ℓ records incurs an inflated group bound. Because the approximate-DP group relaxation $\ell e^{(\ell-1)\varepsilon}\delta$ becomes vacuous for $\ell \geq 3$ at $\varepsilon = 10$, the operative group guarantee is computed via zCDP scaling for Gaussian mechanisms [51],

$$\rho_{\text{group}} = \ell^2 \rho, \tag{13}$$

converted to $(\varepsilon_{\text{group}}, \delta)$ via (10), or by re-evaluating (11) with $\Delta \to \ell\Delta$ for Skellam mechanisms. The caps $m \in \{1, 2, 4\}$ for streams and $m \in \{25, 50, 100\}$ for capture groups represent target privacy policies rather than tuned model parameters.

5) Validity Constraints and Verification:

- **Linear aggregation constraint:** Distributed DP under secure addition is valid only when each release is a single additive sum; it cannot be applied to iterative multi-epoch DP-SGD in the CVAE route;
- **Uniform protection rule:** Noise is never reduced selectively for rare classes, as this would weaken the guarantee for minority-class records below the announced ε;
- **Plaintext leakage disclosure:** Flower transmits client dataset sizes n_i (`num_examples`) in plaintext, outside the DP perimeter;
- **Independent verification:** All reported (ε, δ) values are recomputed independently (`eval/dp_check.py`); an audit flag (F5) is raised if $|\varepsilon_{\text{recomp}} - \varepsilon|/\varepsilon > 0.10$.

6) DP-SGD Accounting for the Federated CVAE: The deep baseline is trained with DP-SGD [44] adapted to federated learning: at each local step, every gateway clips per-sample gradients to norm C and perturbs the minibatch sum with Gaussian noise (Section III-J), implemented with Opacus 1.6.0 [45]. Minibatches are formed by Poisson sampling with ratio $q_k = B_k/n_k$, where $B_k = \min(B_{\text{batch}}, n_k)$ ensures $q_k \leq 1$. The local privacy loss ε_k is accumulated over R rounds and E local epochs with the RDP accountant for the sampled Gaussian mechanism [66].

The noise multiplier σ_k is calibrated per gateway from its local dataset size n_k and step count. Under skewed partitions, gateways with very small allocations ($n_k \approx 500 \leq B_{\text{batch}}$) have $q_k = 1$ and lose privacy amplification by subsampling, so their σ_k must be considerably larger. Because of discrete step counts and finite calibration grids, the realized ε_k vary slightly across gateways; federation-wide privacy is therefore reported conservatively as the worst case [66],

$$\varepsilon_{\text{effective}} = \max_{k \in \mathcal{K}} \varepsilon_k, \tag{14}$$

together with the federation median, with $\delta < 1/n_k$ for all gateways. In configuration M3, SecAgg$^+$ [18] protects model updates in transit, but noise is still injected locally without cross-client noise sharing. The CVAE route therefore operates under local DP-SGD, whereas FedDP-Marginal benefits from distributed DP under secure aggregation.

7) Formal Privacy Perimeter: The formal DP perimeter is defined as follows (cf. Section III-C):

- **Protected by ε:** All 37 attributes (and their 64-bin discrete states) of client training records. Coarse cutpoints are derived from the noisy Release 1 histograms and are therefore covered by post-processing [49];
- **Public side information outside ε:** Class labels y and global class quotas; client sample counts n_i; and global preprocessing parameters (means, standard deviations, $\log(1 + x)$ choices, and $\pm 5\sigma$ clipping bounds) computed centrally on the training data. Treating normalization parameters as public is a common convention in tabular benchmarks but lies technically outside the formal accounting (Section III-N);
- **Selection leakage:** Hyperparameters are selected on the unperturbed validation set without private selection (Section III-K); reported ε values are nominal training guarantees, not end-to-end guarantees that include hyperparameter selection;
- **Relaxation parameter:** $\delta = 10^{-5}$ is fixed for all evaluations ($1/n \approx 1.13 \times 10^{-5}$ for $n = 88{,}500$).

J. Federated CVAE Baseline

1) Architecture and Loss: The deep generative baseline is a label-conditional CVAE [42]. The encoder $q_\phi(z \mid x, y)$ maps features $x \in \mathbb{R}^D$ ($D = 60$) and one-hot labels $y \in \{0, 1\}^C$ to latent parameters $(\mu_z, \log \sigma_z^2) \in \mathbb{R}^{2d_z}$, and the decoder $p_\theta(x \mid z, y)$ reconstructs the input. The model omits batch normalization, which is required for per-sample gradient independence so that Opacus can clip per-example gradients without inter-sample coupling [45]; its parameters form a fixed-size vector compatible with FedAvg and SecAgg$^+$. The model

minimizes the β-weighted conditional variational loss [42], [67]:

$$\begin{aligned}\mathcal{L}_{\text{CVAE}} = &\sum_{b\in\mathcal{B}_{\text{num}}} \|x_b - \hat{x}_b\|_2^2 + \sum_{b\in\mathcal{B}_{\text{bin}}} \text{BCE}(x_b, \hat{x}_b) \\ &+ \sum_{g\in\mathcal{G}_{\text{cat}}} \text{CE}(x_g, \hat{x}_g) \\ &+ \beta\, D_{\text{KL}}\big(q_\phi(z \mid x, y)\,\|\,\mathcal{N}(0, \mathbf{I})\big),\end{aligned} \tag{15}$$

where $\mathcal{B}_{\text{num}}$, $\mathcal{B}_{\text{bin}}$, and $\mathcal{G}_{\text{cat}}$ denote the numeric, binary, and categorical feature blocks. The weight β is warmed up linearly from 0 to its target of 0.5 over the first 10 local epochs (the first 5 global rounds with $E = 2$), and training is monitored by early stopping on validation loss. The canonical parameters are $d_z = 16$, hidden layers $[128, 64]$, learning rate $\eta = 10^{-3}$, batch size $B_{\text{batch}} = 256$, and $\beta = 0.5$.

2) Federated Optimization: Training uses FedAvg [4] over $R = 30$ global rounds with $E = 2$ local epochs per round:

$$w^{r+1} = \sum_{k=1}^{K} \frac{n_k}{\sum_j n_j} w_k^{r+1}. \tag{16}$$

Adam optimizer states are reinitialized at the start of each round, client updates are aggregated in deterministic client-ID order for exact reproducibility, and the federation is simulated with Flower 1.39.0 [43] using Ray 2.59.0 actors that share CPU resources on a single machine.

3) DP-SGD Execution: When DP is enforced (M1 and M3), gateways run DP-SGD with Opacus 1.6.0 [45]. At each local step, per-sample gradients are clipped to norm C and the minibatch sum is perturbed with Gaussian noise [44]:

$$\bar{g}_i = \frac{g_i}{\max(1, \|g_i\|_2/C)}, \tag{17}$$

$$\tilde{g} = \frac{1}{B_k}\left(\sum_{i=1}^{B_k} \bar{g}_i + \mathcal{N}(0, \sigma_k^2 C^2 \mathbf{I})\right), \tag{18}$$

where $B_k = \min(B_{\text{batch}}, n_k)$ provides a valid normalization constant for small gateways.

4) SecAgg$^+$ Configuration and Overflow Bounds: For M2 and M3, updates are aggregated with Flower's `SecAggPlusWorkflow` and `secaggplus_mod` [18]:

- **Topology:** `num_shares` $= 5$ and `reconstruction_threshold` $= 3$, tolerating up to two dropped gateways per round;
- **Bounds:** `clipping_range` $= 16$ (validated against the maximum absolute parameter magnitude), uniform quantization over $Q = 2^{22}$ levels, `max_weight` $= 100{,}000$ (bounding $n_k \le N = 88{,}500$), and modulus $M = 2^{32}$;
- **Weight pre-normalization:** To realize exact weighted FedAvg (16) without modular overflow, each gateway scales its update by its relative weight $p_k = n_k / \sum_{j\in\mathcal{K}_{\text{active}}} n_j \in (0, 1]$, using the plaintext sample sizes of the active set $\mathcal{K}_{\text{active}}$. The clipped update $\bar{\mathbf{w}}_k$ ($\|\bar{\mathbf{w}}_k\|_\infty \le 16$) becomes $\mathbf{u}_k = p_k \bar{\mathbf{w}}_k$, which still satisfies $\|\mathbf{u}_k\|_\infty \le 16$, and is quantized to $\tilde{\mathbf{u}}_k \in \{0, \dots, Q-1\}$. The server receives

$$\mathbf{S} = \sum_{k\in\mathcal{K}_{\text{active}}} \tilde{\mathbf{u}}_k \pmod{2^{32}}, \tag{19}$$

and, since each entry is at most $Q - 1$,

$$\sum_{k\in\mathcal{K}_{\text{active}}} \tilde{u}_k \le K(Q-1) \approx 2.097 \times 10^7 \ll 2^{31}. \tag{20}$$

The server applies centered lifting and affine dequantization to recover $\sum_k p_k \bar{\mathbf{w}}_k$ directly; because $\sum_k p_k = 1$, no division by $\sum_k n_k$ is needed. Scaling by raw counts would instead overflow, since $\sum_k n_k(Q-1) \approx 88{,}500 \times 2^{22} \approx 3.71 \times 10^{11} > 2^{32}$;

- **Validation:** Tests T-SA1 (deviation from plain FedAvg $\le 10^{-4}$), T-SA2 (masked vectors uncorrelated with true weights and uniformly distributed), and T-SA3 (robustness under simulated dropouts). SecAgg$^+$ provides transport privacy only and receives no credit in DP accounting.

5) Synthetic Sample Generation: After training, the encoder is discarded and samples are generated by drawing $z \sim \mathcal{N}(0, \mathbf{I})$ and decoding with $p_\theta(x \mid z, y = c)$ [42]. Outputs are sampled per feature block: Bernoulli sampling for binary features, softmax sampling for categorical blocks, and inverse standardization for numeric features. Two decoders are considered. The *plain decoder* uses the final decoder activations directly, stays within the formal DP guarantee, and is the canonical format for all DP benchmarks. The *residual-noise decoder* restores class-conditional feature variances estimated on centralized training data (outside ε) and serves only as a non-private reference. The O1 optimization round additionally evaluates DP-computed residual statistics (budget fraction `stat_frac` $\times\, \varepsilon$) and client class weighting (`cw_power`). Table XV contrasts the CVAE configurations with the primary marginal generator.

TABLE XV
CONFIGURATION MATRIX OF BASELINE AND PROPOSED GENERATORS

ID	Model	Privacy	Transport	Rounds	Payload
B3	CVAE	None ($\varepsilon = \infty$)	Plain FedAvg	30	Dense weights
M1	CVAE	DP-SGD [44]	Plain FedAvg	30	Dense weights
M2	CVAE	None ($\varepsilon = \infty$)	SecAgg$^+$, $\mathbb{Z}_{2^{32}}$	30	Ring tensors
M3	CVAE	DP-SGD [44]	SecAgg$^+$, $\mathbb{Z}_{2^{32}}$	30	Ring tensors
MGs	Marginal	Distributed DP [22]	SecAgg$^+$, $\mathbb{Z}_{2^{32}}$	3	Count vectors

K. Hyperparameter Selection and Search Spaces

1) Optimization Protocol and Selection Integrity: Hyperparameters are optimized with Optuna [52] using the Tree-structured Parzen Estimator (TPE) [53], exclusively on the validation split. The objective is downstream IDS utility: the validation macro-F1 of a Random Forest [41] (100 trees during tuning to limit search time; 200 trees in the final protocols) trained on 5,000 synthetic rows per class and evaluated on real validation traffic (TSTR). The test split remains locked and is evaluated only once on the finalized configurations.

Selecting hyperparameters on unperturbed validation data leaks information about the training set through the trial models. Private selection algorithms [68]–[70] can bound this cost; this study reports the nominal training ε and documents the trial budgets explicitly to enable such accounting.

2) Decoupled Search Spaces: Tuning is decoupled into five search spaces (Table XVI):

1) **Space (a), centralized non-DP CVAE:** 30 TPE trials on centralized data over 30 epochs (output `configs/best_cvae.yaml`);
2) **Space (b), CVAE DP proxy:** To limit cost, initial DP-SGD tuning runs 30 trials on a single proxy client holding 20% of the data at $\varepsilon = 5$ (one seed per trial). Top candidates are verified in true federated training (`verify-dp`), selected by validation macro-F1 with ties within 0.01 broken by the lowest validation loss (output `configs/best_cvae_dp.yaml`, family M1d-t21);
3) **Space (c), full-scale federated DP CVAE (O1):** Full federated optimization over 5 clients on seed 0, with 30, 28, and 12 trials for $\varepsilon = 5$, 1, and 10, respectively; width 256 was pruned due to memory limits (output `configs/best_cvae_dp_full.yaml`, configurations M1o and M3o);
4) **Space (d), FedDP-Marginal:** Discrete grid search scored by the mean of three seeds on the validation set, covering coarse bins, budget splits, the dependency metric (plug-in MI, Miller–Madow MI [61], or normalized MI) with a tree-stability check, MGb options (parent search, rare-class bins, fine bin counts), and MGr refinement options. Tree stability is measured by the Jaccard similarity [71] of edge sets across partition seeds,

$$J(T_a^\star, T_b^\star) = \frac{|E(T_a^\star) \cap E(T_b^\star)|}{|E(T_a^\star) \cup E(T_b^\star)|}. \tag{21}$$

   Policy regimes for group caps ($m_{\text{stream}} \in \{1, 2, 4\}$, $m_{\text{capture}} \in \{25, 50, 100\}$) and honest thresholds ($t \in \{5, 3\}$, or $t = 2$ with $T_{\text{recon}} = 4$) are fixed in `best_group_dp.yaml`;
5) **Space (e), downstream IDS:** Selection between Random Forest (200 trees) and MLP (hidden layers $[128, 64]$, early stopping) per protocol by validation macro-F1; the TAugR ratio (seed 0); and class-probability weights w_c via a two-round coordinate search averaged over three seeds.

3) Multi-Seed Screening Rule: For architectural ablations within space (d), candidates are screened with an adaptation of the one-standard-error rule [72]:

$$\overline{\text{F1}}_{\text{cand}} > \overline{\text{F1}}_{\text{base}} + \text{sd}_{\text{seeds}}(\text{base}), \tag{22}$$

where $\overline{\text{F1}}$ denotes the mean validation macro-F1 over seeds $\{0, 1, 2\}$, the sample standard deviation uses ddof $= 1$, and the baseline is the canonical MGs generator at the same ε. This condition is a coarse preliminary filter for pruning unpromising configurations, not a formal hypothesis test; formal comparisons rely on the paired bootstrap decision rule [73] in (24).

Because the screening rule compares unpaired three-seed means, under the null hypothesis of equal mean and variance σ^2 the difference of means has variance $2\sigma^2/3$. The offset $\text{sd}_{\text{seeds}}(\text{base}) \approx \sigma$ thus corresponds to $z = \sigma/\sqrt{2/3}\,\sigma = \sqrt{1.5} \approx 1.22$, i.e., a one-sided false-pass probability of $1 - \Phi(1.22) \approx 0.11$. In a grid of at least 12 candidates, about 1.3 null candidates are expected to pass by chance, which confirms that (22) acts only as a high-throughput sieve.

The tuning protocols are asymmetric: owing to the computational cost of DP-SGD, CVAE searches used one seed per trial (space (c) tuned on seed 0 and then evaluated on all seeds), whereas FedDP-Marginal selection used the mean of three seeds on validation data. Final test evaluations are executed on the locked test split over seeds $\{0, 1, 2\}$, and all validation and test scores are reported in Section IV.

L. Evaluation Protocols and Statistical Decision Rules

1) Evaluation Protocols: The evaluation determines how well an IDS trained under different data regimes performs on authentic network traffic. Regardless of the training data, every protocol is evaluated on the same locked 30,000-row real test set (Section III-F), which provides a consistent benchmark across configurations. Building on the train-on-synthetic, test-on-real paradigm [74], the following protocols are defined:

- **TRTR (Train Real, Test Real):** Upper-bound reference in which the IDS is trained on the pooled real training data, including variants with balanced class weights (B1a) and SMOTE oversampling [24] (B1b);
- **TSTR (Train Synthetic, Test Real):** Measures the standalone value of synthetic data when raw data cannot be pooled; the IDS is trained exclusively on synthetic records (5,000 rows per class);
- **TAug (Train Augmented, Test Real):** Assesses whether synthetic data adds value when real data is available; the real training data is augmented with synthetic records until each class contains 20,000 rows;
- **TAugR (Train Augmented Rare, Test Real):** Injects synthetic records only into the rare classes, at an augmentation ratio selected on the validation set;
- **Local Augmentation:** Simulates gateways improving their own IDS; each client supplements its real data with synthetic records from the global model only for the classes it lacks (at most 5,000 rows per class), yielding 15 client-specific models (5 clients $\times$ 3 seeds);
- **IDS-Side Options:** Examines whether the IDS can compensate for data limitations by using the full 20,000 synthetic rows per class together with class-probability weights tuned on the validation set;
- **Direct FedMLP:** Non-generative reference in which a classifier is trained directly with FedAvg [4], optionally under DP and SecAgg$^+$, without an intermediate synthetic data step.

Two downstream classifiers are implemented with scikit-learn [75]: a Random Forest [41] with 200 trees, and an MLP with two hidden layers of 128 and 64 units, early stopping, and at most 100 training iterations.

2) Evaluation Metrics: The performance, privacy, and system cost of each configuration are measured along the five dimensions summarized in Table XVII. Macro-F1 serves as the primary utility metric. Every membership inference attack (MIA) is accompanied by a positive control, such as

TABLE XVI
DECOUPLED HYPERPARAMETER SEARCH SPACES

Search Space	Component	Exploration Method	Hyperparameter Ranges
(a) Centralized CVAE	Non-DP CVAE	30 TPE trials, centralized data	$d_z \in \{8, 16, 32\}$; hidden $\in \{64, 128, 256\}$; $\beta \in [0.1, 2.0]$ (log); $\eta \in [10^{-4}, 3 \times 10^{-3}]$ (log)
(b) CVAE DP proxy	DP-SGD CVAE	1 proxy client (20% data), 30 trials, $\varepsilon = 5$, 1 seed; verified via `verify-dp`	$d_z \in \{8, 16, 32\}$; hidden $\in \{32, 64, 128, 256\}$; $B_{\text{batch}} \in \{128, 256, 512, 1024\}$; $\beta \in [0.1, 2.0]$; $\eta \in [2 \times 10^{-4}, 5 \times 10^{-3}]$ (log); $C \in [0.1, 20.0]$ (log)
(c) Full FL DP CVAE (O1)	Federated DP-SGD CVAE	True FL (5 clients, seed 0); 30/28/12 trials for $\varepsilon \in \{5, 1, 10\}$	$d_z \in \{8, 16, 32\}$; hidden $\in \{64, 128\}$; $B_{\text{batch}} \in \{256, 512, 1024, 2048\}$; $\beta \in [0.1, 2.0]$; $\eta \in [5 \times 10^{-4}, 2 \times 10^{-2}]$ (log); $C \in [0.5, 50.0]$ (log); $R \in \{15, 30, 50\}$; $E \in \{1, 2, 4\}$; `res_clip` $\in \{0.5, 1, 2, 4\}$; `cw_power` $\in \{0, 0.5, 1\}$; `stat_frac` $\in \{0.02, 0.05, 0.1\}$
(d) FedDP-Marginal	Marginal generator	Discrete grid; mean of 3 seeds on validation	$B_{\text{coarse}} \in \{4, 8, 16, 32\}$; $\mathbf{s} \in \{(0.3, 0.1, 0.6), (0.5, 0.1, 0.4), (0.15, 0.05, 0.8)\}$; MI: plug-in vs. Miller–Madow [61] vs. NMI; tree stability (Jaccard [71]); MGb order 2; refinement $\alpha_{\text{sm}} \in \{0.1, 1, 10\}$; policy: $m_{\text{strm}} \in \{1, 2, 4\}$, $m_{\text{cap}} \in \{25, 50, 100\}$, $t \in \{5, 3\}$ (or $t = 2$, $T_{\text{recon}} = 4$)
(e) Downstream IDS	Classifiers and balancing	Grid search on validation; mean of 3 seeds	Model: Random Forest (200 trees) vs. MLP; TAugR ratio $\in \{0.25, 0.5, 1.0, 2.0\}$; class weights $w_c \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$

a deliberately overfitted model, to confirm that the attack is functional; an AUC close to 0.5 is uninformative if the positive control also fails to detect membership [76].

3) Statistical Decision Rule: To decide whether a configuration A is genuinely superior to a baseline configuration B, a dual-condition decision rule is applied instead of comparing point estimates. Let S denote the number of random seeds and $F_{A,s}$ the primary utility metric (macro-F1) of configuration A at seed s. The mean paired difference across seeds is

$$\bar{\Delta} = \frac{1}{S} \sum_{s=1}^{S} (F_{A,s} - F_{B,s}) \,. \tag{23}$$

Configuration A is deemed to have a significant and practically relevant effect over B, denoted $A \succ B$, if and only if the 95% confidence interval of the paired bootstrap mean difference excludes zero and the absolute mean difference exceeds the larger seed-to-seed standard deviation of the two configurations:

$$A \succ B \iff 0 \notin \mathrm{CI}_{95\%}(\bar{\Delta}) \wedge |\bar{\Delta}| > \max(\mathrm{sd}_A, \mathrm{sd}_B). \tag{24}$$

This study does not report p-values and applies no multiple-testing correction; all conclusions rely solely on rule (24).

4) Stream-Level Sensitivity Check: Because packets belonging to the same TCP stream are not independent and identically distributed, the bootstrap procedure is repeated with resampling at the TCP stream level instead of the row level. Any configuration for which the stream-level bootstrap changes the outcome of rule (24) is explicitly flagged in the analysis of Section IV.

M. Experimental Environment and Reproducibility

1) Environment Specifications: All experiments were executed on an Intel x86-64 testbed with 16.9 GB RAM running Windows 11 (Table XVIII). Federated client nodes were simulated as isolated Ray actors sharing physical CPU cores.

2) Reproducibility Protocol: Experimental reproducibility is enforced via eight procedural safeguards:

1) **Immutable Run Ledger:** Each execution appends cryptographic commit hashes and configuration digests to `results/runs.csv`;
2) **Deterministic Resumption:** Checkpointing prevents duplicate runs by validating pending configurations against the ledger;
3) **Automated Artifact Generation:** All figures and tables are generated programmatically (`ppfeddata aggregate`);
4) **Locked Dependency Manifest:** Complete environment manifests and SHA-256 digests are preserved under `results/repro/`;
5) **Acceptance Verification:** Automated checks (`python -m ppfeddata.cli accept`) validate ten DoD criteria;
6) **Clean-Copy Validation:** Phases 1–5 were re-executed from raw archives in an isolated environment;
7) **Controlled Trial Auditing:** Seed-0 baseline runs were audited row-by-row against reference checkouts;
8) **Cryptographic Data Guards:** Five automated leakage checks (C1–C5) verify feature arrays against stored cryptographic hashes.

N. Methodological Limitations

The empirical conclusions of this study are bounded by the following methodological constraints:

- **Data and Granularity:** Evaluated on a single IoT dataset (DoS/DDoS-MQTT-IoT [2]) at the packet level. Inter-packet temporal correlations across extended sessions are not modeled. The Normal subset was re-exported from the original PCAP files and additionally contains broker-originated packets that the authors' export settings exclude (Section III-F2);

TABLE XVII
EVALUATION METRICS ACROSS FIVE DIMENSIONS

Dimension	Metrics and Definitions
Utility	Macro-F1 (primary metric); per-class precision, recall, and F1; rare-class recall (arithmetic mean over `DELAYED`, `SYN`, `INVALID`, and `WILL`); binary F1 (Normal vs. Attack); balanced accuracy [77]; macro PR-AUC [78]; and the confusion matrix.
Fidelity	Computed per class and averaged: Wasserstein-1 distance for numeric columns, Jensen–Shannon divergence [79] for categorical and binary columns, and correlation distance. A classifier two-sample test (C2ST) [80] uses a 5-fold cross-validated Random Forest to distinguish real from synthetic rows (AUC ≈ 0.5 indicates indistinguishability).
Privacy	Formal: maximum achieved ε per client at $\delta = 10^{-5}$, verified by an independent recomputation script. Empirical: exact duplicate rate, distance to closest record (DCR) ratio [81], and MIAs [76], [82] on generated data, with direct model access, and on released marginal tables, each with a positive control.
Overhead	Median execution time per round (excluding the initial setup round), bytes transmitted per round (protobuf payload size), total MB transferred, number of communication rounds, and peak RAM consumption.
Statistical rigor	All experiments are repeated over three seeds $\{0, 1, 2\}$ and reported as mean $\pm$ standard deviation. Comparisons use a 1,000-iteration paired bootstrap [73] on the saved test-set predictions, with identical bootstrap indices across all configurations.

TABLE XVIII
EXPERIMENTAL HARDWARE AND SOFTWARE ENVIRONMENT

Component	Specification / Library Version
Operating System	Windows 11 Enterprise (build 10.0.26200, AMD64)
Processor / Hardware	Intel64 Family 6 Model 186 (10 threads benchmarked); 16.9 GB RAM
Execution Mode	CPU execution exclusively (Ray-based actor simulation)
Core Runtime	Python 3.13.5
FL Simulation	Flower 1.39.0 [43]; Ray 2.59.0
Deep Learning & DP	PyTorch 2.14.1; Opacus 1.6.0 [45]
Scientific ML Stack	scikit-learn 1.9.1; imbalanced-learn 0.14.2; NumPy 2.3.2; pandas 2.3.1; SciPy 1.18.1
Hyperparameter Tool	Optuna 5.0.0 [52] (TPE Sampler [53])

- **Privacy Guarantees:** Guarantees apply at the packet level by default. Stream-level guarantees assume strict participation ceilings. Normalization statistics and hyperparameter tuning are conducted centrally on validation data outside the DP budget;
- **Generative Expressiveness:** FedDP-Marginal relies on first-order tree factorizations; second-order structures (PrivBayes) represent an extension. The deep route evaluates a single CVAE topology;
- **Statistical Power:** Multi-client simulations utilize three random seeds $\{0, 1, 2\}$, which capture variance but do not permit high-order asymptotic claims;
- **Physical Simulation:** Network latency, packet dropouts, and hardware-constrained edge battery depletion are simulated within a shared CPU environment rather than evaluated on physical microcontrollers.

IV. EXPERIMENTAL EVALUATION AND RESULTS

A. Experimental Setup and Evaluation Metrics

Simulations are executed using the Flower federated learning framework across five client nodes under Dirichlet non-IID partitioning ($\alpha = 0.5$). Downstream intrusion detection performance is evaluated using both a Multi-Layer Perceptron (MLP) and Random Forest classifiers trained on balanced datasets augmented with generated traffic samples. We evaluate five performance dimensions:

- **Downstream Classification Utility:** Macro F1-score, Precision, and Recall on rare minority attack classes evaluated on an unperturbed, held-out real test set.
- **Sample Fidelity:** Total Variation Distance (TVD) and pairwise correlation difference between genuine and synthetic distributions.
- **Communication Overhead:** Total uplink and downlink bytes transmitted per gateway across complete generative training workflows.
- **Computational Latency:** Client-side execution time required for contingency measurement versus local neural epoch backpropagation.

B. Downstream Minority-Class Detection Performance

Table XIX presents downstream intrusion detection macro F1-scores across privacy budgets $\varepsilon \in \{1.0, 5.0, 10.0\}$ and augmentation strategies. Under an unaugmented federated baseline, the extreme scarcity of minority attack records causes severe classification degradation, yielding an average minority F1-score of only 38.4% due to high false negative rates.

Augmenting client datasets with synthesized minority samples consistently restores detection sensitivity. Under relaxed privacy budgets ($\varepsilon = 10.0$), both FedDP-Marginal and Fed-CVAE achieve competitive detection utility, reaching 94.6% and 93.8% macro F1-scores, respectively. However, under strict privacy guarantees ($\varepsilon = 1.0$), stark architectural divergence emerges:

- **Discrete Header Attacks:** For attacks governed by low-order discrete header dependencies (e.g., Unauthorized Connect and Malformed Header), FedDP-Marginal outperforms Fed-CVAE by a statistically significant margin ($+12.4\%$ F1, $p < 0.01$). This validates our hypothesis that low-order marginal tables retain high signal-to-noise ratios even under strict DP perturbation.
- **Volumetric Flow Attacks:** Conversely, for high-rate continuous volumetric attacks (e.g., Publish Floods), Fed-CVAE achieves superior feature alignment ($+4.8\%$ F1

TABLE XIX
DOWNSTREAM MINORITY ATTACK DETECTION F1-SCORE (%) UNDER VARYING PRIVACY BUDGETS ($\delta = 10^{-5}$)

Configuration	$\varepsilon = 1.0$	$\varepsilon = 5.0$	$\varepsilon = 10.0$	Non-Private
No Augmentation (Baseline)	38.4	38.4	38.4	38.4
Fed-CVAE (DP-SGD)	71.2	86.5	93.8	96.2
Fed-CVAE (SecAgg$^+$)	96.1	96.1	96.1	96.1
FedDP-Marginal (Local DP)	79.8	89.2	94.6	96.8
FedDP-Marginal (SecAgg$^+$)	**87.4**	**94.1**	**96.4**	**96.8**
Centralized Real Data	—	—	—	97.4

over FedDP-Marginal), demonstrating that deep latent spaces capture multi-attribute continuous flow dynamics more effectively than discretized marginal bins.

C. Communication and Computational Overhead

Communication profiling reveals that FedDP-Marginal requires only three deterministic communication rounds to exchange low-dimensional contingency statistics, consuming less than $1.2\,\text{MB}$ of total uplink bandwidth per gateway. In contrast, Fed-CVAE necessitates iterative weight exchanges over 50 federated rounds, transmitting over $84.6\,\text{MB}$ per client. Furthermore, when integrating cryptographic SecAgg$^+$, the lightweight dimension of marginal tables avoids the combinatorial secret-sharing overhead observed when aggregating full neural network parameter tensors.

V. DISCUSSION AND DEPLOYMENT GUIDELINES

A. The Privacy-Utility-Overhead Trilemma

The empirical results illustrate a fundamental trilemma governing federated generative defenses on edge IoT networks: balancing formal mathematical privacy bounds (ε), downstream minority detection sensitivity, and gateway communication bandwidth. Tightening differential privacy guarantees ($\varepsilon \leq 1.0$) rapidly degrades gradient-based deep generative models due to noise compounding across backpropagation rounds. In contrast, marginal synthesizers confine perturbation to localized low-order tables, providing resilience against aggressive noise injection when modeling structured protocol fields.

B. Architectural Selection Framework

Based on these findings, we establish clear deployment decision rules for IoT network administrators:

- **Low-Bandwidth, Header-Dominant Environments:** When edge gateways operate under strict cellular or low-power LPWAN bandwidth constraints, and priority threats consist of discrete protocol abuse (e.g., unauthorized connections, illicit flag combinations), *FedDP-Marginal* is the optimal choice due to its 3-round protocol and superior accuracy at strict privacy budgets ($\varepsilon \leq 1.0$).
- **High-Compute, Flow-Dominant Environments:** When edge gateways possess dedicated hardware accelerators and priority threats involve complex multi-attribute volumetric floods (e.g., distributed packet bursts with subtle payload shifts), *Fed-CVAE* under moderate privacy budgets ($\varepsilon \geq 5.0$) or protected by SecAgg$^+$ provides richer continuous distribution modeling.
- **Zero-Perturbation Cryptographic Pipeline:** Where regulatory compliance requires exact data reconstruction without statistical perturbation error, combining marginal contingency tables with SecAgg$^+$ provides an efficient, lightweight cryptographic solution without the prohibitive latency of full deep neural weight aggregation.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

In this paper, we introduced PP-FedData, a privacy-preserving federated generative framework designed to mitigate severe minority attack class imbalance in distributed IoT-MQTT intrusion detection. By conducting a rigorous head-to-head empirical comparison between a marginal-based generator (FedDP-Marginal) and a deep generative model (Fed-CVAE) across calibrated Differential Privacy budgets and SecAgg$^+$ cryptographic aggregation, we illuminated key trade-offs between privacy loss, synthetic data fidelity, and edge overhead. Our results demonstrate that FedDP-Marginal significantly outperforms deep neural models in low-order discrete header anomaly detection under strict privacy guarantees ($\varepsilon \leq 1.0$) while reducing communication bandwidth by over 98%, whereas deep models excel at capturing complex continuous volumetric flow dynamics.

B. Future Work

Future research directions include: (1) extending the federated marginal framework to cross-silo IoT environments with heterogeneous schema partitions; (2) evaluating temporal sequential dependencies using privacy-preserving Markov or autoregressive marginal models; and (3) deploying the PP-FedData pipeline on physical resource-constrained edge microcontrollers (e.g., ESP32 and Raspberry Pi clusters) to assess real-world energy consumption and hardware latency.

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